









# Contents

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# Chapter 1

## Introduction

Estimating tail risk is challenging because the events of greatest practical interest are precisely those for which the data are scarce. In financial, insurance and operational-risk applications, the target may lie beyond the range where empirical averages or order statistics alone give reliable answers. Extreme-value methods address this by combining intermediate tail information with a heavy-tailed model, trading a small amount of structural assumption for the ability to extrapolate to rarer loss levels. This is the regime studied here, with a focus on extreme expectiles as tail-risk functionals.

Expectiles, introduced by Newey and Powell (1987), are asymmetric least-squares analogues of quantiles. At high levels they respond to both the size and probability of large losses, and they have attractive risk-measure properties, including coherence in the relevant high-level range and elicibility under standard integrability conditions (Bellini et al. 2014; Ziegel 2016; Gneiting 2011). A formal definition is recalled in Section 2.3; here the point is that high-level expectiles are useful tail-risk functionals whose asymptotic behaviour can be related to extreme quantiles. For heavy-tailed distributions with finite mean, extreme expectiles are also asymptotically proportional to extreme quantiles. In the notation used here,,

$$\xi_\tau \sim \psi(\gamma)Q(\tau), \quad \psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}, \quad \tau \uparrow 1,$$

where  $Q$  is the marginal quantile function and  $\gamma$  is the extreme-value index. This bridge is the point at which expectile inference can be connected to the more developed theory of extreme-quantile extrapolation.

The second ingredient is pooling. In many statistical settings the available sample is naturally distributed across several machines, institutions or data sources. If these sources share the same upper-tail target, each local sample contains useful information but none is necessarily ideal on its own. Daouia et al. (2024) develop an optimal weighted-pooling framework for tail-index and extreme-quantile inference, including geometric pooling of local Weissman estimators. Their results identify variance- and AMSE-optimal weights for the pooled extreme-quantile problem and provide estimated-weight and plug-in inference machinery under source conditions. The question addressed here is how much of that structure survives after the target is changed from an extreme quantile to an extreme expectile.

The estimator studied here applies the expectile–quantile bridge after the pooled extreme-quantile extrapolation. For a target level  $\tau_n = 1 - p_n$ , define

$$\hat{\xi}_{\tau_n}^{\text{pool},\star}(\boldsymbol{\nu}, \boldsymbol{\omega}) = \psi\left(\hat{\gamma}_n(\boldsymbol{\nu})\right) \hat{q}_n^{\star}(\tau_n \mid \boldsymbol{\omega}).$$

The vector  $\boldsymbol{\omega}$  weights the geometrically pooled Weissman estimator  $\hat{q}_n^{\star}(\tau_n \mid \boldsymbol{\omega})$ . The vector  $\boldsymbol{\nu}$  weights only the pooled Hill estimator appearing in the outer bridge factor. This separation is important. The diagonal choice  $\boldsymbol{\nu} = \boldsymbol{\omega}$  is a natural prototype, but it is not forced by the construction and is not treated as an optimality principle.

With this estimator in place, the contribution is split into four parts. The first is the estimator itself: a two-weight construction that keeps the pooled-Weissman weights  $\boldsymbol{\omega}$  separate from the bridge-Hill weights  $\boldsymbol{\nu}$ . The second is the exact decomposition of the log error into a pooled-Weissman term, a bridge-Hill term, and a population bridge term. In the finite- $m$  i.i.d. common-marginal setting, the Chapter 3 statement identifies the pooled-Weissman term as the first-order component under the stated bridge-rate condition.



Thus the first-order bias, variance and AMSE criteria concern  $\omega$ , while  $\nu$  remains first-order unidentified.

The third contribution is a conditional transfer of the relevant estimated-weight and plug-in inference machinery from the optimal-pooling framework of Daouia et al. (2024) to the expectile estimator when the original consistency conditions apply. Estimated weights are used only in the pooled-Weissman role. For implementation, the bridge-Hill vector is kept at the deterministic threshold-allocation convention  $\nu_n^k = (k_1/k, \dots, k_m/k)^\top$ .

The fourth contribution is a finite-sample study matched to this theoretical scope. The simulations vary the pooled-Weissman weights, keep the bridge-Hill convention fixed, and assess point estimation, plug-in intervals and decomposition diagnostics. The intervals are treated as diagnostics of the asymptotic plug-in construction, not as calibrated finite-sample procedures. The study is intentionally limited to the estimator and i.i.d. common-marginal design analysed in Chapter 3; non-identical margins, time series, random bridge-Hill weights and lower-order optimality for  $\nu$  are left outside the present scope.

The rest of the work is organised as follows. Chapter 2 recalls the heavy-tail, expectile and optimal-pooling results used as source inputs. Chapter 3 develops the pooled extreme-expectile estimator, its exact decomposition, the deterministic first-order statement, the weight consequences and the estimated-weight inference transfer statement. Chapter 4 reports the finite-sample study for the proposed estimator and plug-in diagnostics. Chapter 5 summarises the main lessons, discusses the limitations of the estimator and inference construction, and outlines directions for further work.

# Chapter 2

## Background

### 2.1 Regular variation and the second-order condition

We consider a real-valued random loss  $X$  that has continuous distribution function  $F$  and tail, or survival, function  $\bar{F}(x) := 1 - F(x)$ . Given an i.i.d. sample  $X_1, \dots, X_n$  from  $F$ , the corresponding order statistics are denoted by  $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$ .

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is de Haan and Ferreira (2006, Appendix B.1).

**Definition 2.1** (Regular variation). A Lebesgue measurable function  $f : (0, \infty) \rightarrow \mathbb{R}$  that is eventually positive is *regularly varying at infinity with index*  $\alpha \in \mathbb{R}$ , written  $f \in \text{RV}_\alpha$ , if for every  $y > 0$ ,

$$\lim_{t \rightarrow \infty} \frac{f(ty)}{f(t)} = y^\alpha.$$

The functions in  $\text{RV}_0$  are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that  $F$  has a Pareto-type tail.

**Condition 2.2** (First-order condition). There exists  $\gamma > 0$  such that  $\bar{F} \in \text{RV}_{-1/\gamma}$ , or

equivalently the *tail quantile function*

$$U(t) := \inf \left\{ x \in \mathbb{R} : \frac{1}{F(x)} \geq t \right\}, \quad t > 1,$$

satisfies  $U \in \text{RV}_\gamma$ .

Throughout, for a continuous distribution function  $F$ , we write

$$Q_X(\tau) := F^\leftarrow(\tau) = \inf \{ x \in \mathbb{R} : F(x) \geq \tau \}, \quad 0 < \tau < 1.$$

For continuous  $F$ , this agrees with  $U(t) = F^\leftarrow(1 - 1/t)$ . Thus

$$Q_X(\tau) = U((1 - \tau)^{-1}), \quad Q_X(1 - p) = U(1/p).$$

[Condition 2.2](#) is the Fréchet max-domain condition with extreme-value index  $\gamma$  (de Haan and Ferreira [2006](#), Theorem 1.2.1); standard examples include Pareto, Burr, Student- $t$ , Fréchet and Fisher–Snedecor  $F$ -law tails (de Haan and Ferreira [2006](#), §2.6). For normal limits we use the standard second-order refinement.

**Condition 2.3** (Second-order condition  $\mathcal{C}_2(\gamma, \rho, A)$ ). There exist  $\gamma > 0$ , a second-order parameter  $\rho \leq 0$ , and a measurable auxiliary function  $A$  of constant sign with  $A(t) \rightarrow 0$  as  $t \rightarrow \infty$ , such that for every  $y > 0$ ,

$$\lim_{t \rightarrow \infty} \frac{1}{A(t)} \left[ \frac{U(ty)}{U(t)} - y^\gamma \right] = y^\gamma \frac{y^\rho - 1}{\rho},$$

where the right-hand side is read as  $y^\gamma \log y$  when  $\rho = 0$ .

The function  $|A|$  is itself regularly varying with index  $\rho$  (de Haan and Ferreira [2006](#), Theorem 2.3.9). Since  $A$  has constant sign eventually, signed ratios  $A(ty)/A(t)$  converge to  $y^\rho$ ; the rate  $A$  is the source of the second-order bias terms below.

**Condition 2.4** (Finite first-moment range). In addition to [Condition 2.2](#), the extreme-

value index satisfies  $\gamma \in (0, 1)$  and the negative part  $X^- = \max(-X, 0)$  satisfies

$$\mathbb{E} X^- < \infty.$$

Under [Condition 2.2](#),  $\gamma < 1$  gives  $\mathbb{E} X^+ < \infty$ ; together with the displayed lower-tail condition this is the finite-mean regime for the quantile-based expectile approximation.

## 2.2 Tail-index and extreme-quantile estimation

We collect the two classical estimators used throughout this work: the Hill estimator of  $\gamma$  and the Weissman extrapolation for extreme quantiles. The textbook reference is de Haan and Ferreira ([2006](#), Ch. 3–4).

### 2.2.1 The Hill estimator

Given an i.i.d. sample  $X_1, \dots, X_n$  from  $F$  and an integer  $k = k_n \in \{1, \dots, n-1\}$ , the *Hill estimator* (Hill [1975](#)) of  $\gamma$  based on the  $k+1$  top order statistics is

$$\hat{\gamma}_n(k) = \frac{1}{k} \sum_{i=1}^k \log \left( \frac{X_{n-i+1:n}}{X_{n-k:n}} \right). \quad (2.1)$$

Since  $U(t) \rightarrow \infty$ ,  $X_{n-k:n} > 0$  with probability tending to one for intermediate  $k$ ; log-based estimators are understood on this event. The consistency result below is de Haan and Ferreira ([2006](#), Theorem 3.2.2); the normal limit is de Haan and Ferreira ([2006](#), Theorem 3.2.5) when  $\rho < 0$ , with the  $\rho = 0$  endpoint covered by the  $m = 1$  specialisation of Daouia et al. ([2024](#), Theorem 1).

**Theorem 2.5.** *Assume [Condition 2.2](#) with  $\gamma > 0$ . Let  $k = k_n \rightarrow \infty$  and  $k/n \rightarrow 0$  as  $n \rightarrow \infty$ . Then  $\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$ . If, in addition,  $\mathcal{C}_2(\gamma, \rho, A)$  holds and  $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$ , then*

$$\sqrt{k} (\hat{\gamma}_n(k) - \gamma) \xrightarrow{d} \mathcal{N} \left( \frac{\lambda}{1 - \rho}, \gamma^2 \right).$$

The last citation is used only to justify the endpoint case  $\rho = 0$ : with  $m = 1$ , the pooled-Hill central limit theorem reduces to the same single-sample Hill statement under the displayed second-order centring. The choice of  $k$  controls the usual bias–variance trade-off; here it is treated as an exogenous tuning sequence in the asymptotic analysis.

### 2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic  $X_{n-k:n}$  — which estimates the quantile  $Q_X(1 - k/n)$  at the intermediate level  $1 - k/n$  — to a quantile at a very-extreme level  $1 - p$  with  $p = p_n \rightarrow 0$  as  $n \rightarrow \infty$ . Plugging the Hill estimator into the first-order identity  $U(ty)/U(t) \rightarrow y^\gamma$  gives the Weissman estimator

$$\hat{q}_n^*(1 - p \mid k) = \left( \frac{k}{np} \right)^{\hat{\gamma}_n(k)} X_{n-k:n}. \quad (2.2)$$

The theorem below, the Hill-estimator specialisation of de Haan and Ferreira (2006, Theorem 4.3.8), covers the far-tail regime  $np = o(k)$ ; the case  $np = O(1)$  is very-extreme in the sense that only finitely many observations are expected beyond the target. Such rows are included by the theorem whenever  $k/(np) \rightarrow \infty$  and  $\sqrt{k}/\log\{k/(np)\} \rightarrow \infty$  both hold.

**Theorem 2.6.** *Assume  $\mathcal{C}_2(\gamma, \rho, A)$  with  $\gamma > 0$  and  $\rho < 0$ . Let  $k = k_n \rightarrow \infty$ ,  $k/n \rightarrow 0$ ,  $p = p_n \rightarrow 0$  with  $k/(np) \rightarrow \infty$  and  $\sqrt{k}/\log(k/(np)) \rightarrow \infty$ . Suppose  $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$ . Then*

$$\frac{\sqrt{k}}{\log(k/(np))} \left( \frac{\hat{q}_n^*(1 - p \mid k)}{Q_X(1 - p)} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1 - \rho}, \gamma^2\right).$$

The log-denominator is the cost of extrapolation. Up to that factor the limit is the Hill limit, which is why Daouia et al. (2024, §2.3) pool Weissman estimators geometrically. This pooled-Weissman law is the source result against which the pooled extreme-expectile estimator in Chapter 3 will be compared.

## 2.3 Expectiles

Expectiles were introduced by Newey and Powell (1987) as asymmetric-least-squares analogues of quantiles. Their role as coherent and elicitable risk measures is developed in Bellini et al. (2014), Ziegel (2016), Gneiting (2011), and Ehm et al. (2016).

### 2.3.1 Definition

Let  $X$  be a real-valued random variable with  $\mathbb{E}|X| < \infty$ , and fix a level  $\tau \in (0, 1)$ . The  $\tau$ -*expectile* of  $X$  is defined as

$$\xi_\tau(X) := \arg \min_{\theta \in \mathbb{R}} \mathbb{E}[\eta_\tau(X - \theta) - \eta_\tau(X)], \quad \eta_\tau(u) := |\tau - \mathbb{1}\{u \leq 0\}| u^2. \quad (2.3)$$

The subtraction of  $\eta_\tau(X)$  makes the criterion finite under  $\mathbb{E}|X| < \infty$  and does not affect the minimiser; at  $\tau = 1/2$  one recovers the mean.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau \mathbb{E}[(X - \xi_\tau(X))_+] = (1 - \tau) \mathbb{E}[(\xi_\tau(X) - X)_+], \quad (2.4)$$

where  $u_+ = \max(u, 0)$ . Thus expectiles balance upper and lower first partial moments, unlike quantiles, which balance tail probabilities.

### 2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

**Proposition 2.7.** *Let  $X$  and  $Y$  be real-valued random variables with  $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$ . For each  $\tau \in (0, 1)$ :*

(i) **Existence and uniqueness.**  $\xi_\tau(X)$  exists and is uniquely defined.

- (ii) **Law invariance.** If  $X$  and  $Y$  have the same distribution, then  $\xi_\tau(X) = \xi_\tau(Y)$ .
- (iii) **Mean recovery.**  $\xi_{1/2}(X) = \mathbb{E} X$ .
- (iv) **Location equivariance.**  $\xi_\tau(X + c) = \xi_\tau(X) + c$  for  $c \in \mathbb{R}$ .
- (v) **Positive homogeneity.**  $\xi_\tau(\lambda X) = \lambda \xi_\tau(X)$  for  $\lambda \geq 0$ .
- (vi) **Sign reflection.**  $\xi_\tau(-X) = -\xi_{1-\tau}(X)$ .
- (vii) **Monotonicity in  $\tau$ .** The map  $\tau \mapsto \xi_\tau(X)$  is nondecreasing on  $(0, 1)$ , and is strictly increasing when  $X$  is nondegenerate, with the boundary limits  $\xi_\tau(X) \rightarrow \text{ess inf}(X)$  as  $\tau \rightarrow 0$  and  $\xi_\tau(X) \rightarrow \text{ess sup}(X)$  as  $\tau \rightarrow 1$  (infinite limits allowed).
- (viii) **Monotonicity in the argument.** If  $X \leq Y$  almost surely, then  $\xi_\tau(X) \leq \xi_\tau(Y)$ .

For losses, these properties give translation equivariance, positive homogeneity and monotonicity. If  $Y$  denotes a financial position and  $\rho_\tau(Y) = \xi_\tau(-Y)$  is the associated capital requirement, then  $\rho_\tau$  is coherent exactly when  $\tau \geq 1/2$  (Bellini et al. 2014). This is a statement about the capital requirement for positions under the loss convention  $-Y$ , not about subadditivity of the map  $X \mapsto \xi_\tau(X)$  for arbitrary signed variables. Expectiles are also elicitable on classes with finite first moment; the asymmetric squared score is strictly consistent under finite second moments, and the general scoring class has Bregman-type and Choquet-type representations (Gneiting 2011; Ehm et al. 2016). These facts motivate high-level expectiles as tail-risk functionals, while the asymptotic theory below uses only (2.4) and the extreme expectile–quantile equivalence.

## 2.4 The high-expectile/high-quantile bridge

Chapter 3 uses expectile theory through the population relationship between a high expectile and the quantile of the same level. The stochastic extreme-quantile part is supplied by the pooled Weissman estimator of Daouia et al. (2024, §2.3), recalled in Section 2.5; the expectile-specific work is to control the deterministic bridge between  $\xi_\tau$

and  $Q_X(\tau)$ . This section therefore records only the bridge statements needed later.

### 2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

**Proposition 2.8** (Asymptotic equivalence; Bellini et al. (2014), Daouia et al. (2019), Corollary 1 with  $p = 2$ ). *Suppose [Condition 2.4](#) holds. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau \uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}. \quad (2.5)$$

The bridge requires only the finite-mean range  $\gamma < 1$  and lower-tail integrability; stronger restrictions used for direct empirical expectile estimation are not part of the quantile-based bridge route. The function  $\psi$  is continuous on  $(0, 1)$ , with  $\psi(\gamma) \rightarrow 1$  as  $\gamma \downarrow 0$  and  $\psi(\gamma) \rightarrow \infty$  as  $\gamma \uparrow 1$ . It is *not* monotone: its logarithmic derivative

$$m(\gamma) := \frac{d}{d\gamma} \log \psi(\gamma) = \frac{1}{1 - \gamma} - \log(\gamma^{-1} - 1) \quad (2.6)$$

changes sign at  $\gamma^* \approx 0.2178$ , where  $\psi$  reaches its minimum. In the Chapter 3 pooled extreme-expectile statement,  $m(\gamma)$  appears as the coefficient of the linear term in the Taylor expansion of the bridge-Hill contribution. At the exceptional value where  $m(\gamma) = 0$ , this linear term vanishes and the expansion starts at quadratic order.

Under the second-order condition  $\mathcal{C}_2(\gamma, \rho, A)$  with  $\gamma \in (0, 1)$  and  $\mathbb{E} X^- < \infty$ , the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia et al. 2020, Proposition 1(i))

$$\begin{aligned} \frac{\xi_\tau(X)}{Q_X(\tau)} &= \psi(\gamma) \left[ 1 + c(\gamma, \rho) A((1 - \tau)^{-1}) + \gamma(\gamma^{-1} - 1)^\gamma \frac{\mathbb{E} X}{Q_X(\tau)} \right. \\ &\quad \left. + o(|A((1 - \tau)^{-1})|) + o(Q_X(\tau)^{-1}) \right], \\ c(\gamma, \rho) &:= \frac{(\gamma^{-1} - 1)^{-\rho}}{1 - \gamma - \rho} + \frac{(\gamma^{-1} - 1)^{-\rho} - 1}{\rho}. \end{aligned} \quad (2.7)$$



At  $\rho = 0$  the second term in  $c(\gamma, \rho)$  is read by continuity, so  $c(\gamma, 0) = m(\gamma)$ . The two visible deterministic pieces in (2.7) are the second-order tail-bias term

$$c(\gamma, \rho)A((1 - \tau)^{-1})$$

and the first-moment bridge term

$$\gamma(\gamma^{-1} - 1)^\gamma \frac{\mathbb{E} X}{Q_X(\tau)}.$$

In [Chapter 3](#) the first is controlled through the second-order tail rate, while the second is controlled on the pooled-Weissman scale by the explicit bridge-rate condition.

### 2.4.2 Relation to earlier expectile estimators

The i.i.d. extreme-expectile literature also studies direct empirical expectiles based on asymmetric least squares, LAWS-type estimators using least asymmetrically weighted squares, quantile-based intermediate plug-in estimators, and Weissman-type expectile extrapolations from an intermediate expectile anchor (Daouia et al. 2018; Daouia et al. 2020; Davison et al. 2023). These results provide context for the use of the expectile–quantile bridge, but they communicate or estimate a different object. The estimator analysed in [Chapter 3](#) performs the extreme extrapolation in the pooled-quantile component and then applies the population bridge at the target level. Consequently, the later argument relies on the first- and second-order bridge statements recorded in this section rather than on a direct empirical-expectile central limit theorem.

## 2.5 The optimal pooling framework

We now recall the pooling framework of Daouia et al. (2024). For compact references in the simulation chapter, this work sometimes abbreviates their pooling theory as DPS. The framework provides the source theory for pooled tail-index and extreme-quantile inference before the expectile–quantile bridge is applied. The framework pools tail-index

and extreme-quantile estimators from  $m$  samples, allowing heterogeneous sizes, second-order properties and cross-sample tail dependence. This general framework is recalled to identify the source objects and notation. It is broader than the setting used later: the main Chapter 3 statement specialises back to independent samples with common margins.

### 2.5.1 Setup

Throughout [Section 2.5](#),  $m$  is fixed while  $n = \sum_{j=1}^m n_j \rightarrow \infty$ . Let  $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,m})^\top$ ,  $i \geq 1$ , be i.i.d. copies of  $\mathbf{X} = (X_1, \dots, X_m)^\top$ . Margin  $j$  is observed for  $i = 1, \dots, n_j$  and has continuous right-heavy-tailed distribution  $F_j$ , tail quantile function  $U_j(t) = F_j^{\leftarrow}(1 - 1/t)$ , and extreme-value index  $\gamma_j > 0$ ; its order statistics are denoted  $X_{1:n_j,j} \leq \dots \leq X_{n_j:n_j,j}$ . We write  $Q_j(\tau) = F_j^{\leftarrow}(\tau)$ , so that  $U_j(t) = Q_j(1 - 1/t)$  for  $t > 1$ .

Tail inference in sample  $j$  uses its top  $k_j + 1$  order statistics, where the effective sample size  $k_j = k_j(n)$  satisfies  $k_j \rightarrow \infty$  and  $k_j/n_j \rightarrow 0$  as  $n \rightarrow \infty$ . The total effective sample size is  $k = \sum_{j=1}^m k_j$ . Assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \rightarrow b_j \in (0, \infty), \quad \frac{k_1}{k_j} \rightarrow c_j \in (0, \infty), \quad (2.8)$$

with  $b_1 = c_1 = 1$ . The marginal  $F_j$  satisfies  $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$  ([Condition 2.3](#)). The analysis is developed under the *common-tail-index* assumption

$$(H_\gamma) \quad \gamma_1 = \dots = \gamma_m = \gamma.$$

Under this assumption, all common- $\gamma$  vector limits below are expressed on the aggregate scale  $\sqrt{k}$ , with the constants  $c_j$  translating the local  $\sqrt{k_j}$  rates.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure ([Daouia et al. 2024](#), §2.1).

**Condition  $\mathcal{J}(\mathbf{R})$  (tail-copula condition).** For each pair  $(j, \ell)$  with  $j \neq \ell$ , there exists a function  $R_{j,\ell} : [0, \infty]^2 \setminus \{(\infty, \infty)\} \rightarrow [0, \infty)$  such that, for every  $(x_j, x_\ell) \in$

$$[0, \infty]^2 \setminus \{(\infty, \infty)\},$$

$$\lim_{s \rightarrow \infty} s \mathbb{P}\left(1 - F_j(X_j) \leq x_j/s, 1 - F_\ell(X_\ell) \leq x_\ell/s\right) = R_{j,\ell}(x_j, x_\ell).$$

The case  $R_{j,\ell} \equiv 0$  is asymptotic tail independence, implied by the independent samples used later in the document.

### 2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator  $\hat{\gamma}_j(k_j)$  as in (2.1); we collect these into the random vector

$$\hat{\gamma}_n = \left(\hat{\gamma}_1(k_1), \dots, \hat{\gamma}_m(k_m)\right)^\top.$$

Given a weight vector  $\omega \in \mathbb{R}^m$  with  $\omega^\top \mathbf{1} = 1$ , the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat{\gamma}_n(\omega) := \omega^\top \hat{\gamma}_n = \sum_{j=1}^m \omega_j \hat{\gamma}_j(k_j). \quad (2.9)$$

The affine constraint  $\omega^\top \mathbf{1} = 1$  ensures consistency for the common  $\gamma$  under  $(H_\gamma)$ .

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

**Theorem 2.9** (Daouia et al. (2024), Theorem 1). *Assume  $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$  for every  $j$ , the tail-copula condition  $\mathcal{J}(\mathbf{R})$ , the proportionality conditions (2.8), and  $\sqrt{k_j} A_j(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$  for every  $j$ . Then*

$$\left(\sqrt{k_1}(\hat{\gamma}_1(k_1) - \gamma_1), \dots, \sqrt{k_m}(\hat{\gamma}_m(k_m) - \gamma_m)\right)^\top \xrightarrow{d} \mathcal{N}(\mathbf{B}, \mathbf{V}),$$

with  $\mathbf{B} = (\lambda_j/(1 - \rho_j))_{j=1}^m$  and  $\mathbf{V}$  the symmetric matrix whose entries are

$$V_{j,\ell} = \begin{cases} \gamma_j^2, & j = \ell, \\ \frac{\gamma_j \gamma_\ell}{\max(b_j, b_\ell) \sqrt{c_j c_\ell}} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so  $V_{\ell,j} = V_{j,\ell}$  for  $\ell > j$ . Suppose, in addition, the common- $\gamma$  condition  $(H_\gamma)$ . Then the same marginal estimators, now viewed on the common  $\sqrt{k}$  scale, satisfy

$$\sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1}) \xrightarrow{d} \mathcal{N}(\mathbf{B}_c, \mathbf{V}_c),$$

where

$$(\mathbf{B}_c)_j = \sqrt{c_j \sum_{i=1}^m c_i^{-1}} \frac{\lambda_j}{1 - \rho_j} \quad j = 1, \dots, m.$$

The covariance matrix  $\mathbf{V}_c$  is given by

$$(\mathbf{V}_c)_{j,\ell} = \left( \sum_{i=1}^m c_i^{-1} \right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \frac{\gamma^2}{\max(b_j, b_\ell)} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell, \end{cases}$$

with  $(\mathbf{V}_c)_{\ell,j} = (\mathbf{V}_c)_{j,\ell}$  for  $\ell > j$ . Consequently, for any  $\boldsymbol{\omega}$  with  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ ,

$$\sqrt{k}(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \mathbf{B}_c, \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}).$$

Here  $b_j$  records the limiting relative sample size and  $c_j$  records the limiting relative threshold size. The vector  $\mathbf{B}_c$  collects the local second-order Hill biases after conversion to the aggregate  $\sqrt{k}$  scale, while  $\mathbf{V}_c$  is the corresponding covariance matrix. In the independent distributed case used in Chapter 3,  $\mathbf{V}_c$  becomes diagonal. For independent samples, all off-diagonal entries vanish and  $\mathbf{V}_c$  becomes the diagonal matrix

$$\mathbf{V}_c = \gamma^2 \left( \sum_{i=1}^m c_i^{-1} \right) \text{diag}(c_1, \dots, c_m). \quad (2.10)$$

Equation (2.10) is the covariance form to be used when the main Chapter 3 statement is

specialised to independent samples.

### 2.5.3 Variance- and AMSE-optimal weights

The relevant weights solve quadratic programmes under the affine constraint  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ ; unless stated otherwise, they need not be nonnegative. In [Chapter 3](#), these formulas are applied only to the pooled-Weissman weights; they do not define or optimise the separate bridge-Hill weights introduced there.

**Proposition 2.10** (Optimal weights; Daouia et al. (2024), Theorem 1; estimated-weight implementation in §2.2 and Corollary 1). *Suppose  $\mathbf{V}_c$  is positive definite. Under the common- $\gamma$  condition ( $H_\gamma$ ):*

(i) *The variance-optimal weights, minimising  $\boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}$  subject to  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ , are*

$$\boldsymbol{\omega}^{\text{Var}} = \frac{\mathbf{V}_c^{-1} \mathbf{1}}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}},$$

*with corresponding asymptotic variance  $\boldsymbol{\omega}^{\text{Var}, \top} \mathbf{V}_c \boldsymbol{\omega}^{\text{Var}} = (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1})^{-1}$ .*

(ii) *The AMSE-optimal weights, minimising  $(\boldsymbol{\omega}^\top \mathbf{B}_c)^2 + \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}$  subject to  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ , are*

$$\boldsymbol{\omega}^{\text{AMSE}} = \frac{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{1} - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{B}_c}{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c)(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}) - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}.$$

In the independent-samples case,  $\mathbf{V}_c$  is diagonal ([Equation \(2.10\)](#)) and the variance-optimal weights reduce to

$$\omega_j^{\text{Var}} = \left( \sum_{i=1}^m c_i^{-1} \right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \quad (2.11)$$

where the equivalence uses  $c_i^{-1} = \lim k_i/k_1$ . The finite-sample distributed version is

$$\tilde{\boldsymbol{\omega}}_n^{\text{Var}} = (k_1/k, \dots, k_m/k)^\top,$$

which is exactly normalised and converges to  $\boldsymbol{\omega}^{\text{Var}}$ .

Feasible weights replace  $\mathbf{V}_c$  and  $\mathbf{B}_c$  by consistent plug-ins (Daouia et al. 2024, §2.2). If  $\hat{\boldsymbol{\omega}}^\top \mathbf{1} = 1$  exactly and  $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$ , then  $\hat{\gamma}_n(\hat{\boldsymbol{\omega}})$  is  $\sqrt{k}$ -equivalent to  $\hat{\gamma}_n(\boldsymbol{\omega})$ ; the difficult practical step is stable estimation of  $\mathbf{V}_c$  and the bias quantities.

#### 2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile  $q(1-p)$  with  $p = p(n) \rightarrow 0$ , each sample contributes a Weissman estimator

$$\hat{q}_j^*(1-p \mid k_j) = \left( \frac{k_j}{n_j p} \right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}.$$

Because Weissman extrapolation is multiplicative, pooling is geometric on the original scale, equivalently linear on the log scale. It requires asymptotic equivalence of the marginal extreme quantiles:

**Condition 2.11** (Tail homoskedasticity ( $\mathcal{H}$ )). For every  $j \neq \ell$ ,  $U_j(t)/U_\ell(t) \rightarrow 1$  as  $t \rightarrow \infty$ .

Equivalently,  $Q_j(1-p)/Q_\ell(1-p) \rightarrow 1$  as  $p \downarrow 0$ . The pooled extreme-quantile estimator of Daouia et al. (2024, §2.3) is

$$\hat{q}_n^*(1-p \mid \boldsymbol{\omega}) := \prod_{j=1}^m \left[ \hat{q}_j^*(1-p \mid k_j) \right]^{\omega_j}. \quad (2.12)$$

Taking logs reduces pooling to weighted sums of the local identities  $\log \hat{q}_j^*(1-p \mid k_j) = \log(k_j/(n_j p)) \hat{\gamma}_j(k_j) + \log X_{n_j - k_j : n_j, j}$ ; so the Hill part is handled by Theorem 2.9; (2.8) gives  $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$ .

**Theorem 2.12** (Daouia et al. (2024), Theorem 2). *Assume the hypotheses of Theorem 2.9, the common- $\gamma$  condition ( $H_\gamma$ ), the tail-homoskedasticity condition ( $\mathcal{H}$ ), and, in addition,  $\rho_j < 0$  for every  $j = 1, \dots, m$ . Let  $p = p(n) \rightarrow 0$  with  $k/(np) \rightarrow \infty$  and  $\sqrt{k}/\log(k/(np)) \rightarrow \infty$ . Then for any  $\boldsymbol{\omega}$  with  $\boldsymbol{\omega}^\top \mathbf{1} = 1$  and any  $j \in \{1, \dots, m\}$ ,*

$$\frac{\sqrt{k}}{\log(k/(np))} \left( \frac{\hat{q}_n^*(1-p \mid \boldsymbol{\omega})}{Q_j(1-p)} - 1 \right) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \mathbf{B}_c, \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}).$$

The tail-homoskedasticity condition and the restrictions  $\rho_j < 0$  are essential for transferring the extreme-quantile result to a common target. They ensure that recentering by any marginal extreme quantile is asymptotically harmless on the Weissman scale. The strict  $\rho_j < 0$  assumption is part of the Weissman theorem. Together with the bias-rate assumptions  $\sqrt{k_j} A_j(n_j/k_j) \rightarrow \lambda_j$  in [Theorem 2.9](#), it also supplies the rate needed to recenter by any marginal  $Q_j(1-p)$  under  $(\mathcal{H})$  (Daouia et al. [2024](#), Supplementary Material, Lemma B.2): for any  $j, \ell$ ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(|A_j(1/p)|) + O(|A_\ell(1/p)|) = o(k^{-1/2}),$$

along the Weissman sequence.

Thus the pooled Weissman and pooled Hill limits share the same mean, variance and optimal weights up to the log-rate factor. This shared structure is the reason [Chapter 3](#) can use the weight criteria of Daouia et al. ([2024](#)) for the pooled-Weissman component.

### 2.5.5 Estimated weights and plug-in objects

The optimal weights above are oracle quantities because they depend on the asymptotic bias and covariance objects. For inference they are replaced either by deterministic allocation weights, such as  $(k_1/k, \dots, k_m/k)^\top$  in the independent distributed case, or by estimated weights based on plug-in estimates of the same objects. The following result records the stability property used in [Chapter 3](#). Estimated weights are used there only in the pooled-Weissman role; the bridge-Hill vector remains deterministic.

**Proposition 2.13** (Estimated pooling weights; Daouia et al. ([2024](#)), Supplementary Material, Theorem A.1; Theorem 2 and Corollary 8). *Assume the common- $\gamma$  conditions of [Theorem 2.9](#). Let  $\hat{\omega}_n$  be a random weight vector satisfying*

$$\hat{\omega}_n^\top \mathbf{1} = 1, \quad \hat{\omega}_n \xrightarrow{\mathbb{P}} \omega,$$

for some deterministic  $\boldsymbol{\omega}$  with  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ . Then

$$\sqrt{k} \left( \hat{\gamma}_n(\hat{\boldsymbol{\omega}}_n) - \gamma \right) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \mathbf{B}_c, \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}).$$

If, in addition, the assumptions of [Theorem 2.12](#) hold, then

$$\frac{\sqrt{k}}{\log(k/(np))} \left( \frac{\hat{q}_n^*(1-p \mid \hat{\boldsymbol{\omega}}_n)}{Q_j(1-p)} - 1 \right) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \mathbf{B}_c, \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}), \quad j = 1, \dots, m.$$

In the independent common-distribution distributed case, the same statement holds with the diagonal covariance matrix (2.10) and the distributed bias constants of Daouia et al. (2024, Corollary 5).

The Supplementary Material's Theorem A.1 is the generic estimated-weight transfer: exact affine normalisation and convergence in probability of the weights make the random-weight pooled estimator asymptotically equivalent to the pooled estimator with limiting deterministic weights. Applied to [Theorem 2.9](#), this gives the Hill statement. The Weissman statement is the corresponding random-weight version of [Theorem 2.12](#); in the distributed common-marginal case it is the published Corollary 8 of Daouia et al. (2024).

The next proposition separates the probabilistic consistency of the estimated bias and variance objects from the algebra of applying a chosen weight vector.

**Proposition 2.14** (Plug-in bias and variance). *Let  $\widehat{\mathbf{B}}_{cn}$  and  $\widehat{\mathbf{V}}_{cn}$  be estimators of  $\mathbf{B}_c$  and  $\mathbf{V}_c$  satisfying*

$$\widehat{\mathbf{B}}_{cn} \xrightarrow{\mathbb{P}} \mathbf{B}_c, \quad \widehat{\mathbf{V}}_{cn} \xrightarrow{\mathbb{P}} \mathbf{V}_c,$$

and let  $\hat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$ . Define

$$\hat{B}_{\hat{\boldsymbol{\omega}},n} = \hat{\boldsymbol{\omega}}_n^\top \widehat{\mathbf{B}}_{cn}, \quad \hat{V}_{\hat{\boldsymbol{\omega}},n} = \hat{\boldsymbol{\omega}}_n^\top \widehat{\mathbf{V}}_{cn} \hat{\boldsymbol{\omega}}_n.$$

Then

$$\hat{B}_{\hat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}^\top \mathbf{B}_c, \quad \hat{V}_{\hat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}.$$



All vectors and matrices have fixed dimension. The maps  $(w, b) \mapsto w^\top b$  and  $(w, V) \mapsto w^\top V w$  are continuous, so the result follows from the continuous mapping theorem.

Daouia et al. (2024, §2.2 and Corollary 1) give one way to obtain the consistency assumptions in Proposition 2.14 through estimated second-order and tail-dependence quantities. That construction is not assumption-free. In the source it is stated under:

- $\rho_j < 0$  for all margins;
- the parametric second-order form  $A_j(t) = \gamma \beta_j t^{\rho_j}$ ;
- positive definiteness of the asymptotic covariance matrix;
- consistency of the  $\beta_j$  estimators;
- $(\hat{\rho}_j - \rho_j) \log n_j = o_P(1)$ .

In the common-distribution distributed case, their Corollaries 6–7 give the corresponding deterministic variance allocation and plug-in AMSE constructions under the analogous common second-order parameter consistency assumptions.

The log-scale interval for the expectile estimator in Chapter 3 is derived from the transferred plug-in centring and variance objects after the expectile decomposition.

### 2.5.6 Distributed inference as a special case

**Case used in Chapter 3.** In the independent common-distribution distributed setting, the samples are i.i.d. subdivisions of one common population. Then:

- the common- $\gamma$  condition  $(H_\gamma)$  and the tail-homoskedasticity condition  $(\mathcal{H})$  are trivially satisfied;
- the tail copula  $R_{j,\ell}$  between any pair of distinct samples vanishes identically (independence implies tail independence);
- $V_c$  in Theorem 2.9 becomes the diagonal matrix (2.10);

- the variance-optimal weights of Proposition 2.10 reduce to  $\omega_j^{\text{Var}} \propto k_j$  as in (2.11).

Under the stronger balanced-allocation assumptions studied by Daouia et al. (2024, Corollary 6, Theorem 3 and Corollary 8), the variance-optimally pooled Hill and pooled Weissman estimators match the corresponding unfeasible estimators built from the entire combined sample at first order. This oracle equivalence is useful benchmark context for distributed inference. The main Chapter 3 statement uses the more direct consequence needed for the expectile problem: in the independent common-distribution case, the pooled-Weissman bias and variance objects are the diagonal distributed objects displayed above.

Table 2.1: Roadmap of assumptions used in the pooled extreme-expectile chapter.

| Layer                | Main conditions                                                                                                                                                                                                                             | Role in Chapter 3                                                                                                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Heavy-tail inputs    | $\bar{F} \in \text{RV}_{-1/\gamma}$ and $\mathcal{C}_2(\gamma, \rho, A)$ , with $\rho < 0$ for Weissman limits                                                                                                                              | Provide Hill and pooled-Weissman centring, variance and bias rates.                                                            |
| Expectile target     | $0 < \gamma < 1$ , $\mathbb{E} X^- < \infty$ , and eventual upper-tail positivity                                                                                                                                                           | Ensure a finite high expectile, the bridge expansion, and well-defined log Weissman terms with probability tending to one.     |
| Distributed design   | Fixed $m$ , common continuous i.i.d. margins, $n_1/n_j \rightarrow b_j$ , $k_1/k_j \rightarrow c_j$ , $k \rightarrow \infty$ , and $k/n \rightarrow 0$                                                                                      | Make the target common across machines and reduce the pooling objects to the diagonal distributed bias and variance forms.     |
| Extreme target       | $p_n \rightarrow 0$ , $k/(np_n) \rightarrow \infty$ , $\sqrt{k}/\ell_n \rightarrow \infty$ , where $\ell_n = \log\{k/(np_n)\}$                                                                                                              | Define the pooled-Weissman extrapolation scale.                                                                                |
| Bridge negligibility | $\eta_n = \sqrt{k}/\{\ell_n Q(1 - p_n)\} \rightarrow 0$                                                                                                                                                                                     | Makes the population bridge term lower order on the pooled-Weissman scale.                                                     |
| Weights              | Deterministic affine $(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)$ for the base statement; random $\hat{\boldsymbol{\omega}}_n$ only when exactly normalised and $\hat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$ | Separates bridge-Hill weights from pooled-Weissman weights and restricts the random-weight transfer to $\boldsymbol{\omega}$ . |
| Plug-in inference    | Consistency of the selected DPS plug-in centring and variance objects, plus $V_{\boldsymbol{\omega}} > 0$                                                                                                                                   | Allows studentisation and the log-scale interval after the expectile decomposition.                                            |

# Chapter 3

## Pooled extreme expectiles

This chapter develops the pooled extreme-expectile estimator studied here. We work in the finite- $m$  distributed i.i.d. setting of [Section 2.5.6](#): each machine contains observations from the same continuous heavy-tailed loss distribution, and the target is the common marginal expectile  $\xi_{\tau_n}$ . In this setting the population target is unambiguous, and the pooled extreme-quantile component can be handled through the distributed pooled-Weissman theorem of Daouia et al. ([2024](#), Corollary 8).

The construction combines the geometrically pooled Weissman estimator with the asymptotic bridge between high expectiles and high quantiles. This chapter identifies, through an exact log decomposition, which terms remain visible on the pooled-Weissman scale. Under the bridge-rate condition imposed below, the first-order law is inherited from the pooled extreme-quantile component. Accordingly, first-order weight optimisation concerns the pooled-Weissman weights  $\omega$ . The general heavy-tail background allows  $\gamma > 0$ , but the expectile target is finite only in the finite-mean range. All expectile statements below therefore work under  $0 < \gamma < 1$ , together with the lower-tail moment condition in [Condition 2.4](#).

### 3.1 Estimator and exact decomposition

Let  $p_n = 1 - \tau_n$ ,  $n = \sum_{j=1}^m n_j$ ,  $k = \sum_{j=1}^m k_j$ , and

$$\ell_n = \log\left(\frac{k}{np_n}\right).$$

For each machine  $j$ , let  $\hat{\gamma}_j(k_j)$  be the local Hill estimator and

$$\hat{q}_j^*(1 - p_n \mid k_j) = \left(\frac{k_j}{n_j p_n}\right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}$$

be the local Weissman estimator. For an affine weight vector  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ , the geometrically pooled Weissman estimator is

$$\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}) = \prod_{j=1}^m \left[ \hat{q}_j^*(1 - p_n \mid k_j) \right]^{\omega_j}.$$

Since  $\tau_n = 1 - p_n$ , the notation  $1 - p_n$  and  $\tau_n$  refer to the same target level; the formulas below use  $1 - p_n$  when emphasising the tail probability and  $\tau_n$  when emphasising the expectile target. For another affine weight vector  $\boldsymbol{\nu}^\top \mathbf{1} = 1$ , define the pooled Hill estimator used in the outer bridge by

$$\hat{\gamma}_n(\boldsymbol{\nu}) = \sum_{j=1}^m \nu_j \hat{\gamma}_j(k_j).$$

The two-weight pooled extreme-expectile estimator is

$$\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}, \boldsymbol{\omega}) = \psi(\hat{\gamma}_n(\boldsymbol{\nu})) \hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}), \quad \psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}. \quad (3.1)$$

The vector  $\boldsymbol{\omega}$  weights the pooled-Weissman quantile estimator. The vector  $\boldsymbol{\nu}$  weights only the Hill estimator appearing in the outer expectile bridge. The diagonal construction  $\boldsymbol{\nu} = \boldsymbol{\omega}$  is a special case, not a first-order necessity. All distributed estimators considered later are special cases of (3.1).

**Lemma 3.1** (Exact log decomposition). *Let  $Q(1 - p_n)$  denote the common marginal*

quantile and let  $\xi_{\tau_n}$  denote the corresponding expectile. Then

$$\log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}, \boldsymbol{\omega})}{\xi_{\tau_n}} = A_n(\boldsymbol{\omega}) + B_n(\boldsymbol{\nu}) - C_n, \quad (3.2)$$

where

$$A_n(\boldsymbol{\omega}) = \log \frac{\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega})}{Q(1 - p_n)}, \quad B_n(\boldsymbol{\nu}) = \log \psi(\hat{\gamma}_n(\boldsymbol{\nu})) - \log \psi(\gamma),$$

and

$$C_n = \log \frac{\xi_{\tau_n}}{\psi(\gamma)Q(1 - p_n)}.$$

*Verification.* By the definition of the estimator,

$$\begin{aligned} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}, \boldsymbol{\omega})}{\xi_{\tau_n}} &= \log \frac{\psi(\hat{\gamma}_n(\boldsymbol{\nu}))\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega})}{\xi_{\tau_n}} \\ &= \log \psi(\hat{\gamma}_n(\boldsymbol{\nu})) + \log \hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}) - \log \xi_{\tau_n}. \end{aligned}$$

Insert and subtract the two deterministic quantities  $\log Q(1 - p_n)$  and  $\log \psi(\gamma)$ . This gives

$$\begin{aligned} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}, \boldsymbol{\omega})}{\xi_{\tau_n}} &= \{\log \hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}) - \log Q(1 - p_n)\} \\ &\quad + \{\log \psi(\hat{\gamma}_n(\boldsymbol{\nu})) - \log \psi(\gamma)\} \\ &\quad + \{\log \psi(\gamma) + \log Q(1 - p_n) - \log \xi_{\tau_n}\}. \end{aligned}$$

The first bracket is  $A_n(\boldsymbol{\omega})$ , the second bracket is  $B_n(\boldsymbol{\nu})$ , and the third bracket equals

$$\log \frac{\psi(\gamma)Q(1 - p_n)}{\xi_{\tau_n}} = -\log \frac{\xi_{\tau_n}}{\psi(\gamma)Q(1 - p_n)} = -C_n.$$

Combining the three brackets yields (3.2). □

## 3.2 A deterministic two-weight conjecture

The following conjecture records the first-order limit under a rate condition that makes the deterministic population gap between expectiles and quantiles negligible on the pooled-Weissman scale.

**Conjecture 3.2** (Pooled extreme expectile, deterministic weights). *Let  $m$  be fixed. Suppose the observations across and within machines are i.i.d. copies of a common continuous loss variable  $X$ . Let  $U(t) = Q(1 - 1/t)$  be the common tail quantile, and assume  $\mathcal{C}_2(\gamma, \rho, A)$  with  $0 < \gamma < 1$  and  $\rho < 0$ . Assume  $\mathbb{E} X^- < \infty$  and  $U(t) > 0$  for all sufficiently large  $t$ .*

*Let  $n = \sum_{j=1}^m n_j$ ,  $k = \sum_{j=1}^m k_j$ , and assume*

$$\frac{n_1}{n_j} \rightarrow b_j \in (0, \infty), \quad \frac{k_1}{k_j} \rightarrow c_j \in (0, \infty),$$

*with  $b_1 = c_1 = 1$ . Suppose*

$$k \rightarrow \infty, \quad \frac{k}{n} \rightarrow 0, \quad \sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}.$$

*Let  $p_n = 1 - \tau_n$ ,  $\ell_n = \log\{k/(np_n)\}$ , and assume the extreme Weissman regime*

$$p_n \rightarrow 0, \quad \frac{k}{np_n} \rightarrow \infty, \quad \frac{\sqrt{k}}{\ell_n} \rightarrow \infty.$$

*Finally impose the expectile-bridge rate condition*

$$\eta_n = \frac{\sqrt{k}}{\ell_n Q(1 - p_n)} \rightarrow 0. \tag{3.3}$$

*This condition makes the deterministic population expectile-quantile bridge error negligible on the pooled-Weissman scale. It is consistent with the usual single-sample quantile-based expectile rate conditions: for example, if at an intermediate anchor  $u_n \uparrow 1$  one has  $\sqrt{k}/Q(u_n) = O(1)$ , and the extreme target satisfies  $1 - p_n \geq u_n$  eventually, then monotonicity of  $Q$  gives  $\sqrt{k}/Q(1 - p_n) = O(1)$ , so  $\eta_n = O(\ell_n^{-1}) = o(1)$  whenever  $\ell_n \rightarrow \infty$ . The inequality  $1 - p_n \geq u_n$  means that the target quantile is no smaller than the intermediate anchor quantile. Let  $\boldsymbol{\nu}_n, \boldsymbol{\omega}_n \in \mathbb{R}^m$  be deterministic weight sequences satisfying exact affine normalisation and finite-dimensional convergence,*

$$\boldsymbol{\nu}_n^\top \mathbf{1} = 1, \quad \boldsymbol{\omega}_n^\top \mathbf{1} = 1, \quad \boldsymbol{\nu}_n \rightarrow \boldsymbol{\nu}, \quad \boldsymbol{\omega}_n \rightarrow \boldsymbol{\omega},$$

for finite vectors  $\boldsymbol{\nu}, \boldsymbol{\omega} \in \mathbb{R}^m$ . Fixed deterministic weights are included by taking constant sequences. No non-negativity is imposed unless stated separately. Then the estimator (3.1) satisfies

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}), \quad (3.4)$$

where

$$B_{\boldsymbol{\omega}} = \frac{\lambda}{1-\rho} \sum_{j=1}^m d_j^{\rho} \omega_j, \quad V_{\boldsymbol{\omega}} = \gamma^2 \left( \sum_{j=1}^m c_j^{-1} \right) \left( \sum_{j=1}^m c_j \omega_j^2 \right),$$

and

$$d_j = \frac{c_j}{b_j} \frac{\sum_{i=1}^m c_i^{-1}}{\sum_{i=1}^m b_i^{-1}}.$$

The displayed centring and variance depend on the limiting pooled-Weissman weights  $\boldsymbol{\omega}$ , but not on the limiting bridge-Hill weights  $\boldsymbol{\nu}$ . Moreover,

$$\frac{\sqrt{k}}{\ell_n} \left( \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} - 1 \right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}). \quad (3.5)$$

*Supporting argument.* Write  $s_n = \sqrt{k}/\ell_n$ . By the exact decomposition (3.2),

$$\log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} = A_n(\boldsymbol{\omega}_n) + B_n(\boldsymbol{\nu}_n) - C_n.$$

We now control the three terms on the common scale  $s_n$ . All logarithmic and Taylor expansions below are understood on a common high-probability event. This event has three parts:

- the relevant high order statistics and local Weissman estimators are positive;
- the relative errors to which  $\log(1+x)$  is applied are bounded away from  $-1$ ;
- $\hat{\gamma}_n(\boldsymbol{\nu}_n)$  lies in a fixed compact subinterval of  $(0, 1)$ .

The eventual upper-tail positivity assumption gives the first positivity property. The pooled-Weissman relative-error limit, the expectile-quantile bridge expansion and the pooled-Hill consistency checks below give the remaining parts.

**The pooled-Weissman term**  $A_n(\boldsymbol{\omega}_n)$ . The pooled-Weissman result of Daouia et al. (2024, Corollary 8) controls this term. Specifically, the random-weight version recalled in Proposition 2.13, applied to the exactly normalised deterministic sequence  $\boldsymbol{\omega}_n \rightarrow \boldsymbol{\omega}$ , gives

$$\frac{\sqrt{k}}{\ell_n} \left( \frac{\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}_n)}{Q(1 - p_n)} - 1 \right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Set

$$x_n = \frac{\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}_n)}{Q(1 - p_n)} - 1.$$

The condition  $s_n = \sqrt{k}/\ell_n \rightarrow \infty$  implies  $x_n = O_P(s_n^{-1}) = o_P(1)$ . Hence the relative-error statement can be converted to log scale:

$$\log(1 + x_n) = x_n + O_P(x_n^2)$$

and

$$s_n \{\log(1 + x_n) - x_n\} = O_P(s_n^{-1}) = o_P(1).$$

Therefore

$$s_n A_n(\boldsymbol{\omega}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}). \quad (3.6)$$

**The bridge-Hill term**  $B_n(\boldsymbol{\nu}_n)$ . This term enters only through the smooth map

$$g(x) = \log \psi(x) = -x \log(x^{-1} - 1).$$

On every compact subinterval of  $(0, 1)$ ,  $g$  is twice continuously differentiable, with

$$g'(x) = m(x) = \frac{1}{1 - x} - \log(x^{-1} - 1), \quad g''(x) = \frac{1}{(1 - x)^2} + \frac{1}{1 - x} + \frac{1}{x}.$$

First check that the local Hill estimators are in the usual pooled-Hill regime. The aggregate assumptions imply  $k_j \rightarrow \infty$ ,  $k_j/n_j \rightarrow 0$ , and

$$\frac{k_j}{k} \rightarrow \frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}}, \quad \frac{n_j}{n} \rightarrow \frac{b_j^{-1}}{\sum_{i=1}^m b_i^{-1}}.$$



Consequently,

$$\frac{n_j/k_j}{n/k} \rightarrow d_j = \frac{c_j}{b_j} \frac{\sum_{i=1}^m c_i^{-1}}{\sum_{i=1}^m b_i^{-1}},$$

and regular variation of  $A$  gives

$$\frac{A(n_j/k_j)}{A(n/k)} \rightarrow d_j^\rho$$

by the uniform convergence theorem for regularly varying functions. Hence

$$\sqrt{k_j} A(n_j/k_j) = \sqrt{\frac{k_j}{k}} \sqrt{k} A(n/k) \frac{A(n_j/k_j)}{A(n/k)} \rightarrow \lambda \left( \frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}} \right)^{1/2} d_j^\rho.$$

Thus the assumptions of [Conjecture 3.2](#) meet the local assumptions of [Theorem 2.9](#). Each local Hill estimator has the required  $k_j$ -scale second-order centring, and the vector CLT applies. In particular,

$$\sqrt{k}\{\hat{\gamma}_n - \gamma \mathbf{1}\} = O_P(1).$$

The use of a deterministic triangular bridge-weight sequence does not change this order. Since  $\boldsymbol{\nu}_n \rightarrow \boldsymbol{\nu}$ , the sequence  $(\boldsymbol{\nu}_n)$  is bounded. Using the exact affine normalisation  $\boldsymbol{\nu}_n^\top \mathbf{1} = 1$ ,

$$\sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} = \boldsymbol{\nu}_n^\top \sqrt{k}\{\hat{\gamma}_n - \gamma \mathbf{1}\} = O_P(1).$$

Therefore  $\hat{\gamma}_n(\boldsymbol{\nu}_n) \xrightarrow{\mathbb{P}} \gamma$ . Choose a deterministic compact interval  $K \subset (0, 1)$  containing  $\gamma$  in its interior, and define the random interval

$$I_n = \{\gamma + t\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} : 0 \leq t \leq 1\}.$$

Then  $\Pr(I_n \subset K) \rightarrow 1$ . On this event, Taylor's theorem with remainder bounded uniformly over  $K$  gives

$$B_n(\boldsymbol{\nu}_n) = m(\gamma)\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} + O_P(\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\}^2) = m(\gamma)\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} + O_P(k^{-1}).$$

Consequently

$$s_n B_n(\boldsymbol{\nu}_n) = \frac{m(\gamma)}{\ell_n} \sqrt{k} \{ \hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma \} + O_P\left(\frac{1}{\ell_n \sqrt{k}}\right) = o_P(1), \quad (3.7)$$

because  $\ell_n \rightarrow \infty$ , which follows from  $k/(np_n) \rightarrow \infty$ . If  $m(\gamma) = 0$ , the displayed conclusion follows from the quadratic remainder alone.

**The population bridge term  $C_n$ .** Put

$$a_n = A(1/p_n), \quad b_n = Q(1 - p_n)^{-1}, \quad \alpha_\gamma = \gamma(\gamma^{-1} - 1)^\gamma, \quad \mu = \mathbb{E} X.$$

The second-order expectile-quantile expansion (2.7) gives, after dividing by  $\psi(\gamma)$ ,

$$\Delta_n := \frac{\xi_{\tau_n}}{\psi(\gamma)Q(1 - p_n)} - 1 = c(\gamma, \rho)a_n + \alpha_\gamma \mu b_n + o(|a_n|) + o(b_n).$$

Since  $\Delta_n \rightarrow 0$ ,

$$C_n = \log(1 + \Delta_n) = \Delta_n + O(\Delta_n^2).$$

It remains only to compare the linear terms and the quadratic remainder from the logarithm with  $s_n^{-1}$ . For the second-order tail term, use the regular variation of  $|A|$  and Potter bounds. Since  $|A| \in \text{RV}_\rho$  with  $\rho < 0$ , for any  $\varepsilon \in (0, -\rho)$ ,

$$\left| \frac{A(1/p_n)}{A(n/k)} \right| \leq C_\varepsilon \left( \frac{k}{np_n} \right)^{\rho + \varepsilon}$$

eventually. The right-hand side tends to zero because  $(1/p_n)/(n/k) = k/(np_n) \rightarrow \infty$ . Together with  $\sqrt{k}A(n/k) \rightarrow \lambda$ , this yields  $s_n a_n \rightarrow 0$ . For the mean term, the imposed bridge-rate condition (3.3) is exactly

$$s_n b_n \rightarrow 0.$$

Thus all linear terms in  $C_n$  are negligible after multiplication by  $s_n$ .

The quadratic remainder from the logarithm is negligible as well. Since  $s_n|a_n| \rightarrow 0$ ,  $s_nb_n \rightarrow 0$ , and  $|a_n| + b_n \rightarrow 0$ ,

$$s_n(|a_n| + b_n)^2 = \{s_n(|a_n| + b_n)\}\{|a_n| + b_n\} \rightarrow 0.$$

Hence

$$s_n C_n \rightarrow 0. \tag{3.8}$$

**Putting the terms together.** Since (3.8) is deterministic convergence, it also implies  $s_n C_n = o_P(1)$ . Hence the exact decomposition can be written as

$$s_n \log \frac{\widehat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} = s_n A_n(\boldsymbol{\omega}_n) + o_P(1).$$

Together with (3.6), Slutsky's theorem gives (3.4).

**The relative-error version.** Define

$$L_n = \log \frac{\widehat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}}.$$

The log limit implies  $L_n = O_P(s_n^{-1}) = o_P(1)$ . Therefore

$$e^{L_n} - 1 = L_n + O_P(L_n^2)$$

and

$$s_n \{e^{L_n} - 1 - L_n\} = O_P(s_n^{-1}) = o_P(1).$$

This yields (3.5). □

*Remark 3.3* (What the conjecture does and does not identify). The limit in [Conjecture 3.2](#) depends on  $\boldsymbol{\omega}$ , the limiting geometric pooled-Weissman weights, but not on  $\boldsymbol{\nu}$ , the limiting bridge-Hill weights. Thus  $\boldsymbol{\nu}$  is first-order unidentified in the present regime. The diagonal estimator  $\boldsymbol{\nu}_n = \boldsymbol{\omega}_n$  is a permissible deterministic convention when the common deterministic

sequence satisfies the conjecture's convergence condition, and the practical convention used for plug-in inference below is

$$\boldsymbol{\nu}_n^k = \left( \frac{k_1}{k}, \dots, \frac{k_m}{k} \right)^\top.$$

Neither convention is an optimality result for  $\boldsymbol{\nu}$ . Any optimal choice of bridge-Hill weights would require a lower-order theory not developed here.

### 3.3 Weight consequences

The preceding conjecture determines the relevant first-order weight criterion. The leading bias and variance coincide with those of the pooled-Weissman component of Daouia et al. (2024, Corollary 8) in the distributed common-marginal case. Define

$$\mathbf{B}_c^{\text{dist}} = \frac{\lambda}{1-\rho} (d_1^\rho, \dots, d_m^\rho)^\top, \quad \mathbf{V}_c^{\text{dist}} = \gamma^2 \left( \sum_{i=1}^m c_i^{-1} \right) \text{diag}(c_1, \dots, c_m).$$

Then  $B_\omega = \boldsymbol{\omega}^\top \mathbf{B}_c^{\text{dist}}$  and  $V_\omega = \boldsymbol{\omega}^\top \mathbf{V}_c^{\text{dist}} \boldsymbol{\omega}$ .

**Conjecture 3.4** (First-order criteria for  $\boldsymbol{\omega}$ ). *Under the assumptions of Conjecture 3.2, the first-order variance criterion for the pooled extreme-expectile estimator is*

$$\boldsymbol{\omega}^\top \mathbf{V}_c^{\text{dist}} \boldsymbol{\omega},$$

and the first-order AMSE criterion is

$$(\boldsymbol{\omega}^\top \mathbf{B}_c^{\text{dist}})^2 + \boldsymbol{\omega}^\top \mathbf{V}_c^{\text{dist}} \boldsymbol{\omega},$$

both under the affine constraint  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ . Hence the deterministic/oracle variance- and AMSE-optimal formulas of Daouia et al. (2024) apply to  $\boldsymbol{\omega}$ . These criteria concern  $\boldsymbol{\omega}$  only; the bridge-Hill weights  $\boldsymbol{\nu}$  do not enter because  $B_n(\boldsymbol{\nu}_n)$  is lower order under (3.3). In

particular,

$$\boldsymbol{\omega}^{\text{Var}} = \frac{(\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{1}}{\mathbf{1}^\top (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{1}},$$

so, in the independent distributed case,

$$\omega_j^{\text{Var}} = \frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}}, \quad j = 1, \dots, m.$$

If  $\mathbf{V}_c^{\text{dist}}$  is positive definite, the AMSE-optimal weights are

$$\boldsymbol{\omega}^{\text{AMSE}} = \frac{(1 + (\mathbf{B}_c^{\text{dist}})^\top (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{B}_c^{\text{dist}}) (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{1} - (\mathbf{1}^\top (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{B}_c^{\text{dist}}) (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{B}_c^{\text{dist}}}{(1 + (\mathbf{B}_c^{\text{dist}})^\top (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{B}_c^{\text{dist}}) (\mathbf{1}^\top (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{1}) - (\mathbf{1}^\top (\mathbf{V}_c^{\text{dist}})^{-1} \mathbf{B}_c^{\text{dist}})^2}.$$

No first-order variance or AMSE criterion for  $\boldsymbol{\nu}$  is identified by [Conjecture 3.2](#).

Supporting argument. By [Conjecture 3.2](#),

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_\omega, V_\omega).$$

Thus the scaled first-order mean-square criterion is  $B_\omega^2 + V_\omega$ , and the unscaled criterion differs only by the common factor  $\ell_n^2/k$ . The same criterion applies to relative error by [\(3.5\)](#). Substituting  $B_\omega = \boldsymbol{\omega}^\top \mathbf{B}_c^{\text{dist}}$  and  $V_\omega = \boldsymbol{\omega}^\top \mathbf{V}_c^{\text{dist}} \boldsymbol{\omega}$  gives exactly the quadratic criteria recalled in [Proposition 2.10](#). The variance and AMSE formulas follow from that result.  $\square$

*Remark 3.5* (Centralised benchmark equivalence). The centralised estimator used as a benchmark in the finite-sample study is also covered by the same decomposition. Let  $\hat{q}_n^{\text{cent},*}(1 - p_n)$  and  $\hat{\gamma}_n^{\text{cent}}$  denote the Weissman and Hill estimators computed from the combined sample with aggregate threshold  $k$ , and set

$$\hat{\xi}_{\tau_n}^{\text{cent},*} = \psi(\hat{\gamma}_n^{\text{cent}}) \hat{q}_n^{\text{cent},*}(1 - p_n).$$

Under the additional benchmark-equivalence conditions of Daouia et al. ([2024](#), Corollary 6, Theorem 3 and Corollary 8), recalled in [Section 2.5.6](#), suppose the variance-weighted

pooled-Weissman estimator satisfies

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}_n^{\text{Var}})}{\hat{q}_n^{\text{cent},*}(1 - p_n)} = o_P(1).$$

Then, for any deterministic bridge-Hill sequence  $\boldsymbol{\nu}_n$  satisfying the conditions of [Conjecture 3.2](#),

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n^{\text{Var}})}{\hat{\xi}_{\tau_n}^{\text{cent},*}} = o_P(1).$$

Indeed, with  $g = \log \psi$ ,

$$\log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n^{\text{Var}})}{\hat{\xi}_{\tau_n}^{\text{cent},*}} = \log \frac{\hat{q}_n^*(1 - p_n \mid \boldsymbol{\omega}_n^{\text{Var}})}{\hat{q}_n^{\text{cent},*}(1 - p_n)} + \{g(\hat{\gamma}_n(\boldsymbol{\nu}_n)) - g(\hat{\gamma}_n^{\text{cent}})\}.$$

The pooled-Hill and single-sample Hill CLTs give  $\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma = O_P(k^{-1/2})$  and  $\hat{\gamma}_n^{\text{cent}} - \gamma = O_P(k^{-1/2})$ . On the same compact event used in the proof of [Conjecture 3.2](#), Taylor's theorem gives

$$g(\hat{\gamma}_n(\boldsymbol{\nu}_n)) - g(\hat{\gamma}_n^{\text{cent}}) = O_P(k^{-1/2}), \quad \frac{\sqrt{k}}{\ell_n} \{g(\hat{\gamma}_n(\boldsymbol{\nu}_n)) - g(\hat{\gamma}_n^{\text{cent}})\} = o_P(1).$$

Thus the expectile benchmark equivalence is inherited from the pooled-Weissman benchmark equivalence. This is a narrow statement for the variance-weighted benchmark setup just described, not a claim for arbitrary  $\boldsymbol{\omega}$  choices or arbitrary heterogeneous-threshold designs.

### 3.4 Estimated pooled-Weissman weights and inference

The deterministic conjecture is the base statement. Estimated weights are introduced only in the pooled-Weissman role  $\boldsymbol{\omega}$ , where Daouia et al. (2024, Supplementary Material, Theorem A.1, Theorem 2 and Corollary 8) provide random-weight quantile limits under their own consistency conditions. The bridge-Hill vector remains deterministic in this chapter. When the estimated weights or plug-in bias and variance objects are taken

from the constructions of Daouia et al. (2024, §2.2 and Corollary 1, Corollaries 6–7), the consistency assumptions recalled in Section 2.5.5 are imposed.

The outside input needed here is the random-weight version of the pooled-Weissman limit recorded in Proposition 2.13: exact affine normalisation  $\hat{\omega}_n^\top \mathbf{1} = 1$  and convergence  $\hat{\omega}_n \xrightarrow{\mathbb{P}} \omega$  give the same first-order quantile limit as the deterministic vector  $\omega$ . The rest of the argument is a transfer through the exact decomposition.

**Conjecture 3.6** (Estimated- $\omega$  transfer). *Assume the conditions of Conjecture 3.2, with the deterministic pooled-Weissman weight sequence in the estimator replaced by a random  $\hat{\omega}_n$  satisfying the estimated-weight conditions of Proposition 2.13, with limiting vector  $\omega$ . Let  $\nu_n$  be any deterministic bridge-Hill weight sequence satisfying the exact affine normalisation and convergence conditions of Conjecture 3.2; in particular one may take  $\nu_n = \nu_n^k = (k_1/k, \dots, k_m/k)^\top$ . Then*

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\nu_n, \hat{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_\omega, V_\omega).$$

*The same limit holds on relative-error scale.*

*Supporting argument.* The exact decomposition becomes

$$\log \frac{\hat{\xi}_{\tau_n}^{\text{pool},*}(\nu_n, \hat{\omega}_n)}{\xi_{\tau_n}} = A_n(\hat{\omega}_n) + B_n(\nu_n) - C_n.$$

Proposition 2.13 gives the random-weight relative-error limit for the pooled-Weissman component. With

$$x_n(\hat{\omega}_n) = \frac{\hat{q}_n^*(1 - p_n \mid \hat{\omega}_n)}{Q(1 - p_n)} - 1,$$

this gives

$$\frac{\sqrt{k}}{\ell_n} x_n(\hat{\omega}_n) \xrightarrow{d} \mathcal{N}(B_\omega, V_\omega).$$

Since  $\sqrt{k}/\ell_n \rightarrow \infty$ ,  $x_n(\hat{\omega}_n) = O_P(\ell_n/\sqrt{k}) = o_P(1)$ , and the relative-to-log Taylor step

used in the supporting argument for [Conjecture 3.2](#) gives

$$\frac{\sqrt{k}}{\ell_n} A_n(\hat{\omega}_n) \xrightarrow{d} \mathcal{N}(B_\omega, V_\omega).$$

The bridge-Hill term is unchanged because  $\boldsymbol{\nu}_n$  is deterministic, exactly normalised, and convergent. The argument for (3.7) therefore applies verbatim. The population bridge term  $C_n$  is deterministic and is still controlled by (3.8). Slutsky's theorem gives the log limit, and the final exponential Taylor step in [Conjecture 3.2](#) gives the relative-error version.  $\square$

Thus any weight construction covered by [Proposition 2.13](#) transfers in the pooled-Weissman role, including estimated variance- and AMSE-optimal weights when the corresponding assumptions give  $\hat{\omega}_n \xrightarrow{\mathbb{P}} \omega$ . This statement concerns  $\omega$  only. It does not optimise  $\boldsymbol{\nu}$ , and it does not cover a random bridge-Hill convention such as  $\boldsymbol{\nu} = \hat{\omega}_n$ .

**Conjecture 3.7** (Plug-in studentisation and log-scale interval). *Let*

$$\hat{\xi}_{\tau_n}^{\text{plug}} = \hat{\xi}_{\tau_n}^{\text{pool},*}(\boldsymbol{\nu}_n^k, \hat{\omega}_n), \quad \boldsymbol{\nu}_n^k = \left( \frac{k_1}{k}, \dots, \frac{k_m}{k} \right)^\top.$$

*Under the assumptions of [Conjecture 3.6](#), let  $\widehat{\mathbf{B}}_{c_n}^{\text{dist}}$  and  $\widehat{\mathbf{V}}_{c_n}^{\text{dist}}$  be plug-in estimators satisfying*

$$\widehat{\mathbf{B}}_{c_n}^{\text{dist}} \xrightarrow{\mathbb{P}} \mathbf{B}_c^{\text{dist}}, \quad \widehat{\mathbf{V}}_{c_n}^{\text{dist}} \xrightarrow{\mathbb{P}} \mathbf{V}_c^{\text{dist}}.$$

*These consistency assumptions belong to the selected plug-in construction from Daouia et al. (2024, §2.2 and Corollary 1, Corollaries 6–7); they are not consequences of the expectile decomposition alone. No expectile-specific bias or variance estimator is introduced here. Define*

$$\widehat{B}_{\hat{\omega},n} = \hat{\omega}_n^\top \widehat{\mathbf{B}}_{c_n}^{\text{dist}}, \quad \widehat{V}_{\hat{\omega},n} = \hat{\omega}_n^\top \widehat{\mathbf{V}}_{c_n}^{\text{dist}} \hat{\omega}_n,$$



and assume  $V_\omega > 0$ . Then

$$T_n^{\text{plug}} = \frac{\frac{\sqrt{k}}{\ell_n} \log\left(\frac{\hat{\xi}_{\tau_n}^{\text{plug}}}{\xi_{\tau_n}}\right) - \hat{B}_{\hat{\omega},n}}{\sqrt{\hat{V}_{\hat{\omega},n}}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Solving the corresponding studentised inequality for  $\xi_{\tau_n}$  gives a multiplicative interval on the original scale. Let  $z_{1-\alpha/2}$  be the standard normal quantile. On the high-probability event that the local Weissman estimators are positive and  $\hat{V}_{\hat{\omega},n} > 0$ , define

$$I_{n,1-\alpha}^{\text{plug}} = \left[ \hat{\xi}_{\tau_n}^{\text{plug}} \exp\left\{-\frac{\ell_n}{\sqrt{k}} \left(\hat{B}_{\hat{\omega},n} + z_{1-\alpha/2} \sqrt{\hat{V}_{\hat{\omega},n}}\right)\right\}, \right. \\ \left. \hat{\xi}_{\tau_n}^{\text{plug}} \exp\left\{-\frac{\ell_n}{\sqrt{k}} \left(\hat{B}_{\hat{\omega},n} - z_{1-\alpha/2} \sqrt{\hat{V}_{\hat{\omega},n}}\right)\right\} \right].$$

Then

$$\mathbb{P}\{\xi_{\tau_n} \in I_{n,1-\alpha}^{\text{plug}}\} \rightarrow 1 - \alpha.$$

Supporting argument. [Conjecture 3.6](#) gives

$$\frac{\sqrt{k}}{\ell_n} \log\left(\frac{\hat{\xi}_{\tau_n}^{\text{plug}}}{\xi_{\tau_n}}\right) \xrightarrow{d} \mathcal{N}(B_\omega, V_\omega).$$

[Proposition 2.14](#), applied with  $\mathbf{B}_c = \mathbf{B}_c^{\text{dist}}$  and  $\mathbf{V}_c = \mathbf{V}_c^{\text{dist}}$ , gives

$$\hat{B}_{\hat{\omega},n} \xrightarrow{\mathbb{P}} B_\omega, \quad \hat{V}_{\hat{\omega},n} \xrightarrow{\mathbb{P}} V_\omega.$$

The additional expectile terms have already been absorbed in [Conjecture 3.6](#), so the centring and variance objects remain the pooled-Weissman objects supplied by Daouia et al. ([2024](#), §2.2 and Corollary 1, Corollaries 6–7). Together with  $V_\omega > 0$ , the displayed convergences give the standard normal limit by Slutsky's theorem. Since  $\hat{V}_{\hat{\omega},n} \xrightarrow{\mathbb{P}} V_\omega > 0$ , the event  $\hat{V}_{\hat{\omega},n} > 0$  has probability tending to one. Eventual upper-tail positivity also gives positivity of the high order statistics, and hence of all local Weissman estimators, with

probability tending to one. Solving

$$\left|T_n^{\text{plug}}\right| \leq z_{1-\alpha/2}$$

for the unknown  $\xi_{\tau_n}$  gives the displayed multiplicative interval, so its asymptotic coverage follows from the studentised limit.  $\square$

*Remark 3.8* (Scope of the inference conjecture). The interval in [Conjecture 3.7](#) is obtained by inverting the plug-in-centred expectile statistic above. It is therefore a consequence of the transferred expectile limit, rather than a direct restatement of the quantile interval in Daouia et al. ([2024](#)). The centering and variance objects remain those of the pooled-Weissman component because, under  $\eta_n \rightarrow 0$ , the additional expectile-bridge terms are lower order on that scale. The conjecture is consequently conditional on the selected plug-in construction satisfying the consistency statements established by Daouia et al. ([2024](#), §2.2 and Corollary 1, Corollaries 6–7) for  $B_\omega$  and  $V_\omega$ ; it does not make an additional claim about simulations, finite-sample calibration, or optimal bridge-Hill weights.

# Chapter 4

## Finite-sample study

This chapter reports a finite-sample study for the estimator developed in [Chapter 3](#). The purpose is deliberately narrow. The simulations study

$$\hat{\xi}_{\tau}^{\text{pool},*}(\boldsymbol{\nu}_n^k, \boldsymbol{\omega}) = \psi(\hat{\gamma}_n(\boldsymbol{\nu}_n^k)) \hat{q}_n^*(1 - p \mid \boldsymbol{\omega}), \quad \boldsymbol{\nu}_n^k = (k_1/k, \dots, k_m/k)^\top,$$

where  $p = 1 - \tau$  and  $k = \sum_{j=1}^m k_j$ . Thus the bridge-Hill weights are fixed at the deterministic threshold-allocation convention, while the pooled-Weissman weights  $\boldsymbol{\omega}$  vary across the variance and AMSE choices of Daouia et al. (2024) studied below. The role of  $\boldsymbol{\nu}_n^k$  is that of a transparent implementation convention rather than an optimality claim; all weight variation in the study concerns  $\boldsymbol{\omega}$ .

The empirical conclusions should be read as diagnostics for the Chapter 3 first-order argument, not as new mathematical claims. The final run uses 5000 Monte Carlo replications per scenario. The main grid targets  $np = 5$ , while the target-sensitivity block compares  $np \in \{1, 5, 10\}$ . In the representative imbalanced design, the variance weights of Daouia et al. (2024) make the distributed estimator track the centralised benchmark closely for point estimation. Equal pooled-Weissman weights are less efficient under imbalance. In the main equal-threshold grid, AMSE weights are almost indistinguishable from variance weights; in the separate heterogeneous-threshold diagnostic they become more informative,

but can also produce unstable signed weights. The log-scale plug-in intervals have empirical coverage below nominal levels in several models, so the finite-sample evidence supports the estimator more strongly than it supports the nominal first-order interval calibration.

## 4.1 Estimators and implementation

For each machine  $j$ , the local Hill estimator and Weissman quantile estimator are computed using the `HTailIndex` and `extQuantile` routines from the `ExtremeRisks` package (Padoan 2026). The simulation code adds only the study-specific pooling layer: geometric pooling of local Weissman estimates, the map  $\psi$ , the deterministic bridge-Hill convention  $\boldsymbol{\nu}_n^k$ , and the plug-in summary calculations.

The following estimators are recorded. As noted in Section 2.5, *DPS* is used in tables as a compact label for the pooling theory of Daouia et al. (2024).

- *equal*:  $\omega_j = 1/m$ ;
- *DPS variance*:  $\omega_j = k_j/k$ , the independent distributed version of the variance weights of Daouia et al. (2024) in this common-tail design. Since the variance-optimal formula reduces to this deterministic threshold allocation, the oracle and plug-in variance-weight versions coincide;
- *DPS AMSE oracle*: the AMSE formula of Daouia et al. (2024) evaluated with the population second-order parameters of the data-generating model;
- *DPS AMSE plug-in*: the same formula with second-order inputs estimated using the `evt0` package (Belzile et al. 2026);
- *centralised*: the corresponding single-sample estimator computed from the combined sample, used only as a benchmark.

All distributed estimators use the same bridge-Hill vector  $\boldsymbol{\nu}_n^k$ . The plug-in AMSE estimator randomises only  $\boldsymbol{\omega}$ . For exact Pareto models, the second-order bias coefficient is zero in

the design, so the plug-in AMSE construction is treated as source-ineligible rather than as a distinct estimated AMSE procedure. The oracle AMSE row then collapses to the same point estimator as the variance rule. For the centralised benchmark interval, the centring and variance objects are the corresponding single-sample oracle objects obtained by taking  $m = 1$ . Thus dashes in exact-Pareto plug-in AMSE rows mean “not reported under the DPS convention”, not numerical failure or zero coverage.

For each estimator with available centring and variance objects, inference is assessed with the log-scale interval from [Conjecture 3.7](#). In one Monte Carlo replication, if  $\hat{B}_\omega$  and  $\hat{V}_\omega$  denote the selected centring and variance objects from Daouia et al. (2024, §2.2 and Corollary 1, Corollaries 6–7), the interval is

$$\hat{\xi} \left[ \exp \left\{ -\frac{\ell}{\sqrt{k}} (\hat{B}_\omega + z_{0.975} \sqrt{\hat{V}_\omega}) \right\}, \exp \left\{ -\frac{\ell}{\sqrt{k}} (\hat{B}_\omega - z_{0.975} \sqrt{\hat{V}_\omega}) \right\} \right], \quad \ell = \log \left( \frac{k}{np} \right).$$

The interval is included as a diagnostic of the first-order plug-in construction, not as a claim of finite-sample calibration.

## 4.2 Design

The study stays within the i.i.d. common-marginal setting of the Chapter 3 conjectural result. The data-generating models are Pareto with  $\gamma = 0.25$ , Pareto with  $\gamma = 0.60$ , Burr with  $\gamma = 0.25$  and  $\rho = -1$ , Burr with  $\gamma = 0.60$  and  $\rho = -1$ , and the absolute Student distribution with three degrees of freedom. The latter has  $\gamma = 1/3$ . The true expectile  $\xi_\tau$  is computed numerically from the population expectile equation

$$\tau \mathbb{E}[(X - \xi_\tau)_+] = (1 - \tau) \mathbb{E}[(\xi_\tau - X)_+],$$

rather than by the asymptotic expectile–quantile bridge.

The final finite-sample simulation uses 5000 Monte Carlo replications per scenario. The grid has 55 scenarios: 20 main-grid scenarios, 15 threshold-sensitivity scenarios, 15

target-sensitivity scenarios, and 5 heterogeneous-threshold scenarios. The main grid uses  $n = 10000$ ,  $k_j/n_j = 0.05$ ,  $np = 5$ ,  $m \in \{5, 10\}$ , and either balanced allocation or a strong imbalanced allocation with machine proportions increasing geometrically from 1 to 10. The threshold block fixes  $m = 10$  and strong allocation, then varies  $k_j/n_j \in \{0.03, 0.05, 0.10\}$ . The target block fixes the same design and varies  $np \in \{1, 5, 10\}$ . The sensitivity blocks were simulated independently from the main grid, so rows that nominally coincide with the representative design are independent Monte Carlo estimates rather than reused entries.

Table 4.1: Compact summary of the finite-sample design.

| Component                | Specification                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data-generating models   | Pareto ( $\gamma = 0.25$ ), Pareto ( $\gamma = 0.60$ ), Burr ( $\gamma = 0.25, \rho = -1$ ), Burr ( $\gamma = 0.60, \rho = -1$ ), and $ t_3 $ .               |
| Replications             | 5000 per final finite-sample scenario.                                                                                                                        |
| Main grid                | $n = 10000$ , $m \in \{5, 10\}$ , balanced or strong allocation, $k_j/n_j = 0.05$ , and $np = 5$ .                                                            |
| Threshold block          | $m = 10$ , strong allocation, $np = 5$ , and $k_j/n_j \in \{0.03, 0.05, 0.10\}$ .                                                                             |
| Target block             | $m = 10$ , strong allocation, $k_j/n_j = 0.05$ , and $np \in \{1, 5, 10\}$ .                                                                                  |
| Heterogeneous thresholds | $m = 10$ , strong allocation, $n = 10000$ , $np = 5$ , and local threshold fractions decreasing with machine size, rescaled to aggregate $k/n \approx 0.05$ . |

The heterogeneous-threshold block is separate. It keeps  $m = 10$ , strong allocation,  $n = 10000$ , and  $np = 5$ , but assigns larger threshold fractions to smaller machines and smaller threshold fractions to larger machines. The profile is rescaled so the aggregate  $k/n$  remains close to 0.05. This block is included because the main grid uses a common local threshold fraction, in which case  $(k_1/k, \dots, k_m/k)^\top$  and  $(n_1/n, \dots, n_m/n)^\top$  coincide.

Performance is measured by the Monte Carlo mean of the log relative error, the root mean squared log relative error, empirical coverage of the nominal 95% log interval, average log interval width, invalid-estimate rate, and weight-stability diagnostics. The code also records the exact finite-sample decomposition terms

$$A_n(\boldsymbol{\omega}) = \log\{\hat{q}_n^*(1-p \mid \boldsymbol{\omega})/Q(1-p)\}, \quad B_n(\boldsymbol{\nu}) = \log \psi\{\hat{\gamma}_n(\boldsymbol{\nu})\} - \log \psi(\gamma),$$

and

$$C_n = \log\{\xi_\tau/(\psi(\gamma)Q(1-p))\}.$$

These diagnostics connect the Monte Carlo evidence to [Lemma 3.1](#); they are not used to estimate new asymptotic constants. Parenthesised values in the generated tables are Monte Carlo standard errors. For means and rates they use the usual empirical standard-error formulas, while RMSE standard errors use the delta method applied to the Monte Carlo mean of the squared log error.

### 4.3 Point estimation and pooled-Weissman weights

[Table 4.2](#) reports the representative main design with  $m = 10$ , strong sample-size imbalance,  $k_j/n_j = 0.05$ , and  $np = 5$ . The estimator using the variance weights of Daouia et al. (2024) is close to the centralised benchmark across the five models. Its log RMSE is 0.223 versus 0.226 for Burr heavy, 0.070 versus 0.069 for Burr light, 0.204 versus 0.204 for Pareto heavy, 0.090 versus 0.089 for Pareto light, and 0.142 versus 0.143 for the absolute Student model. This is the main positive finite-sample finding: in this grid, the proposed distributed estimator does not pay a large point-estimation penalty relative to the single-sample benchmark.

Equal pooled-Weissman weights are worse under strong imbalance. The effect is most visible in the heavier-tailed models, where the equal-weight log RMSE is 0.245 for Burr heavy and 0.231 for Pareto heavy, compared with 0.223 and 0.204 under the DPS variance weights. The oracle and plug-in AMSE weights are essentially the same as the variance weights in this equal-threshold main design.

This closeness has a direct explanation from the DPS AMSE criterion. In the main grid the threshold fraction  $k_j/n_j$  is common across machines, so the asymptotic constants in [Conjecture 3.4](#) satisfy  $c_j = b_j$  and therefore  $d_j = 1$  for every  $j$ . Hence  $\mathbf{B}_c^{\text{dist}}$  is proportional to  $\mathbf{1}$ . Under the affine constraint  $\boldsymbol{\omega}^\top \mathbf{1} = 1$ , the bias term  $\boldsymbol{\omega}^\top \mathbf{B}_c^{\text{dist}}$  is then the same for every admissible  $\boldsymbol{\omega}$ , and the AMSE criterion differs across weights only through the variance term. Thus the DPS AMSE rule collapses asymptotically to the DPS variance rule in this equal-threshold design. This statement is deliberately local to the main grid; [Table 4.10](#) gives AMSE weights a more discriminating environment by varying the local

threshold fractions.

The blank plug-in AMSE rows for exact Pareto models are the table-level form of the convention described above: the construction is source-ineligible in these rows, although the corresponding oracle AMSE point estimator coincides with the variance rule.

Table 4.2: Point-estimation performance in the representative main design ( $m = 10$ , strong allocation,  $k_j/n_j = 0.05$ ,  $np = 5$ ).

| DGP            | $m$ | Regime | Estimator        | Log bias       | Log RMSE      | Valid rate    |
|----------------|-----|--------|------------------|----------------|---------------|---------------|
| Burr heavy     | 10  | strong | Centralised      | 0.077 (0.003)  | 0.226 (0.002) | 1.000 (0.000) |
| Burr heavy     | 10  | strong | DPS AMSE oracle  | 0.075 (0.003)  | 0.223 (0.002) | 1.000 (0.000) |
| Burr heavy     | 10  | strong | DPS AMSE plug-in | 0.075 (0.003)  | 0.223 (0.002) | 1.000 (0.000) |
| Burr heavy     | 10  | strong | DPS variance     | 0.075 (0.003)  | 0.223 (0.002) | 1.000 (0.000) |
| Burr heavy     | 10  | strong | Equal            | 0.070 (0.003)  | 0.245 (0.003) | 1.000 (0.000) |
| Burr light     | 10  | strong | Centralised      | -0.039 (0.001) | 0.069 (0.001) | 1.000 (0.000) |
| Burr light     | 10  | strong | DPS AMSE oracle  | -0.041 (0.001) | 0.070 (0.001) | 1.000 (0.000) |
| Burr light     | 10  | strong | DPS AMSE plug-in | -0.041 (0.001) | 0.070 (0.001) | 1.000 (0.000) |
| Burr light     | 10  | strong | DPS variance     | -0.041 (0.001) | 0.070 (0.001) | 1.000 (0.000) |
| Burr light     | 10  | strong | Equal            | -0.043 (0.001) | 0.084 (0.001) | 1.000 (0.000) |
| Pareto heavy   | 10  | strong | Centralised      | -0.007 (0.003) | 0.204 (0.002) | 1.000 (0.000) |
| Pareto heavy   | 10  | strong | DPS AMSE oracle  | -0.012 (0.003) | 0.204 (0.002) | 1.000 (0.000) |
| Pareto heavy   | 10  | strong | DPS AMSE plug-in | —              | —             | —             |
| Pareto heavy   | 10  | strong | DPS variance     | -0.012 (0.003) | 0.204 (0.002) | 1.000 (0.000) |
| Pareto heavy   | 10  | strong | Equal            | -0.012 (0.003) | 0.231 (0.002) | 1.000 (0.000) |
| Pareto light   | 10  | strong | Centralised      | -0.069 (0.001) | 0.089 (0.001) | 1.000 (0.000) |
| Pareto light   | 10  | strong | DPS AMSE oracle  | -0.071 (0.001) | 0.090 (0.001) | 1.000 (0.000) |
| Pareto light   | 10  | strong | DPS AMSE plug-in | —              | —             | —             |
| Pareto light   | 10  | strong | DPS variance     | -0.071 (0.001) | 0.090 (0.001) | 1.000 (0.000) |
| Pareto light   | 10  | strong | Equal            | -0.071 (0.001) | 0.100 (0.001) | 1.000 (0.000) |
| Abs. Student-3 | 10  | strong | Centralised      | 0.110 (0.001)  | 0.143 (0.001) | 1.000 (0.000) |
| Abs. Student-3 | 10  | strong | DPS AMSE oracle  | 0.109 (0.001)  | 0.142 (0.001) | 1.000 (0.000) |
| Abs. Student-3 | 10  | strong | DPS AMSE plug-in | 0.109 (0.001)  | 0.142 (0.001) | 1.000 (0.000) |
| Abs. Student-3 | 10  | strong | DPS variance     | 0.109 (0.001)  | 0.142 (0.001) | 1.000 (0.000) |
| Abs. Student-3 | 10  | strong | Equal            | 0.108 (0.002)  | 0.154 (0.001) | 1.000 (0.000) |



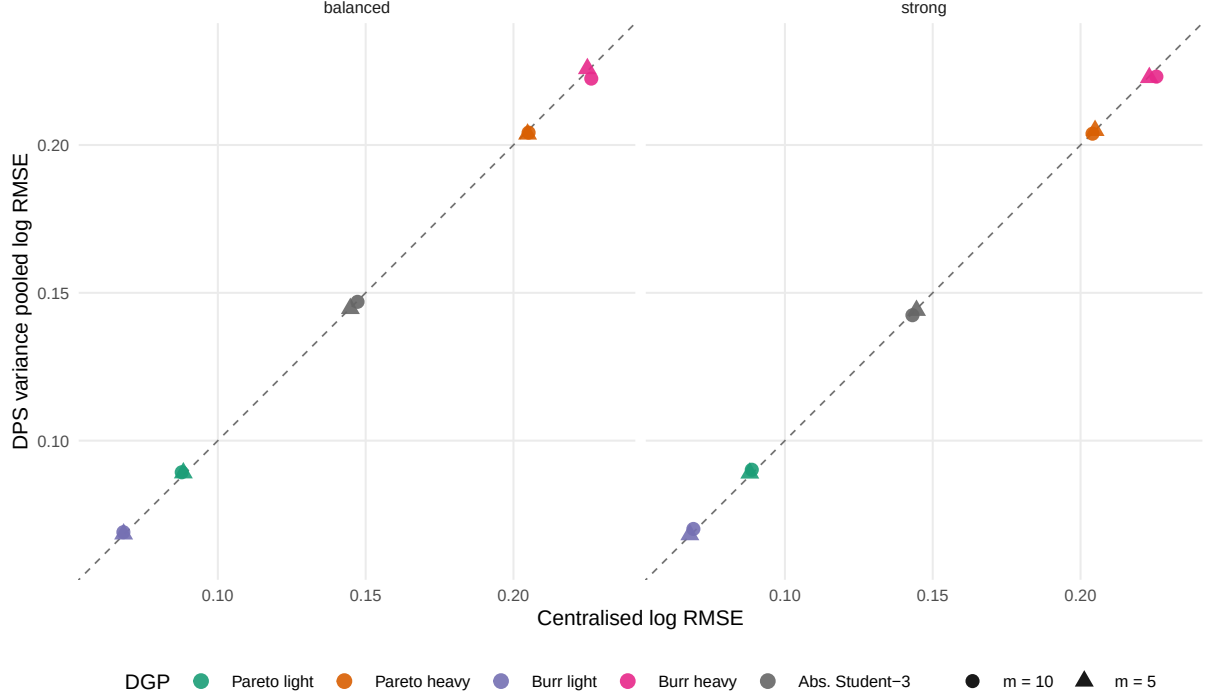


Figure 4.1: Log RMSE of the pooled estimator with DPS variance weights against the centralised benchmark in the main grid. The dashed line is equality.

## 4.4 Deterministic bridge-Hill sensitivity

The main specification fixes the bridge-Hill weights at  $\boldsymbol{\nu}_n^k = (k_1/k, \dots, k_m/k)^\top$ . To check whether this choice has visible finite-sample consequences, Table 4.3 repeats the representative main design for the variance pooled-Weissman weights of Daouia et al. (2024) while replacing  $\boldsymbol{\nu}_n^k$  by two other deterministic choices: equal bridge-Hill weights and sample-allocation weights  $(n_1/n, \dots, n_m/n)^\top$ .

In the representative main design, the threshold and sample-allocation conventions coincide because  $k_j/n_j$  is constant across machines. The table therefore mainly compares this allocation convention against equal bridge-Hill weights. Equal bridge-Hill weights have little effect in the light-tailed rows, but are less favourable in Burr heavy, Pareto heavy, and the absolute Student model. The conclusion is only that the deterministic threshold-allocation choice is not visibly harmful in this main grid.

Table 4.4 makes the comparison more informative by breaking the equality between

threshold and sample-allocation weights. The differences remain modest. In Burr heavy, the sample bridge vector has the lowest log RMSE, 0.231, while  $\nu_n^k$  has slightly higher coverage, 0.744. In the absolute Student model, sample weights again have the lowest log RMSE, while  $\nu_n^k$  remains close. In the light-tailed models, all three bridge-Hill conventions are nearly indistinguishable. These patterns are consistent with the Chapter 3 first-order message:  $\nu$  is first-order unidentified under the chosen scaling, but lower-order finite-sample differences can still be visible.

Table 4.3: Sensitivity to deterministic bridge-Hill weights in the representative main design. The pooled-Weissman weights are fixed at the variance choice of Daouia et al. (2024).

| DGP            | Bridge weights | Log bias       | Log RMSE      | Coverage      |
|----------------|----------------|----------------|---------------|---------------|
| Burr heavy     | Equal          | 0.077 (0.003)  | 0.236 (0.003) | 0.720 (0.006) |
| Burr heavy     | Sample         | 0.075 (0.003)  | 0.223 (0.002) | 0.751 (0.006) |
| Burr heavy     | $\nu^k$        | 0.075 (0.003)  | 0.223 (0.002) | 0.751 (0.006) |
| Burr light     | Equal          | -0.041 (0.001) | 0.070 (0.001) | 0.593 (0.007) |
| Burr light     | Sample         | -0.041 (0.001) | 0.070 (0.001) | 0.591 (0.007) |
| Burr light     | $\nu^k$        | -0.041 (0.001) | 0.070 (0.001) | 0.591 (0.007) |
| Pareto heavy   | Equal          | -0.007 (0.003) | 0.216 (0.002) | —             |
| Pareto heavy   | Sample         | -0.012 (0.003) | 0.204 (0.002) | —             |
| Pareto heavy   | $\nu^k$        | -0.012 (0.003) | 0.204 (0.002) | —             |
| Pareto light   | Equal          | -0.071 (0.001) | 0.090 (0.001) | —             |
| Pareto light   | Sample         | -0.071 (0.001) | 0.090 (0.001) | —             |
| Pareto light   | $\nu^k$        | -0.071 (0.001) | 0.090 (0.001) | —             |
| Abs. Student-3 | Equal          | 0.110 (0.001)  | 0.144 (0.001) | 0.910 (0.004) |
| Abs. Student-3 | Sample         | 0.109 (0.001)  | 0.142 (0.001) | 0.920 (0.004) |
| Abs. Student-3 | $\nu^k$        | 0.109 (0.001)  | 0.142 (0.001) | 0.920 (0.004) |

Table 4.4: Sensitivity to deterministic bridge-Hill weights under heterogeneous local threshold fractions. The aggregate  $k/n$  is kept close to 0.05, and the pooled-Weissman weights are fixed at the variance choice of Daouia et al. (2024).

| DGP            | Bridge weights | Log bias       | Log RMSE      | Coverage      | RMS scaled $B_n$ |
|----------------|----------------|----------------|---------------|---------------|------------------|
| Burr heavy     | Equal          | 0.106 (0.003)  | 0.242 (0.003) | 0.742 (0.006) | 0.554 (0.006)    |
| Burr heavy     | Sample         | 0.090 (0.003)  | 0.231 (0.002) | 0.733 (0.006) | 0.504 (0.006)    |
| Burr heavy     | $\nu^k$        | 0.097 (0.003)  | 0.234 (0.002) | 0.744 (0.006) | 0.501 (0.005)    |
| Burr light     | Equal          | -0.034 (0.001) | 0.067 (0.001) | 0.603 (0.007) | 0.023 (0.000)    |
| Burr light     | Sample         | -0.035 (0.001) | 0.067 (0.001) | 0.599 (0.007) | 0.020 (0.000)    |
| Burr light     | $\nu^k$        | -0.034 (0.001) | 0.067 (0.001) | 0.600 (0.007) | 0.020 (0.000)    |
| Pareto heavy   | Equal          | -0.008 (0.003) | 0.212 (0.002) | –             | 0.415 (0.005)    |
| Pareto heavy   | Sample         | -0.010 (0.003) | 0.210 (0.002) | –             | 0.416 (0.004)    |
| Pareto heavy   | $\nu^k$        | -0.010 (0.003) | 0.208 (0.002) | –             | 0.387 (0.004)    |
| Pareto light   | Equal          | -0.071 (0.001) | 0.091 (0.001) | –             | 0.015 (0.000)    |
| Pareto light   | Sample         | -0.071 (0.001) | 0.091 (0.001) | –             | 0.015 (0.000)    |
| Pareto light   | $\nu^k$        | -0.072 (0.001) | 0.091 (0.001) | –             | 0.014 (0.000)    |
| Abs. Student-3 | Equal          | 0.138 (0.001)  | 0.167 (0.001) | 0.918 (0.004) | 0.275 (0.001)    |
| Abs. Student-3 | Sample         | 0.127 (0.001)  | 0.157 (0.001) | 0.925 (0.004) | 0.222 (0.001)    |
| Abs. Student-3 | $\nu^k$        | 0.132 (0.001)  | 0.162 (0.001) | 0.923 (0.004) | 0.245 (0.001)    |

## 4.5 Plug-in interval performance

Table 4.5 reports nominal 95% log intervals in the representative main design. For non-Pareto distributed rows, the table reports the plug-in construction satisfying the DPS source conditions. For exact Pareto models, the distributed plug-in interval rows are omitted because the plug-in second-order construction is source-ineligible; the centralised oracle benchmark is still shown. The same distinction is made visually in Figure 4.2: exact-Pareto distributed interval cells are marked by dashes rather than interpreted as zero coverage.

The intervals are not uniformly calibrated at this sample size. The distributed Burr heavy intervals cover about 75% of replications, the Burr light intervals cover about 59%, and the absolute Student intervals cover about 92%. The centralised oracle benchmark also has empirical coverage below the nominal level in every model, with coverage between 0.708 and 0.770. Hence the problem is not simply that the sample has been split across machines. The finite sample is also feeling first-order centring, variance, and bridge approximations. In the light-tailed rows,  $\eta_n$  is not numerically small, so these designs should be read as stress tests of the bridge approximation rather than as settings where the bridge-rate

condition is already well reflected.

Figure 4.3 gives a targeted diagnostic for this undercoverage. It uses oracle centring and variance objects to compare coverage against both the exact population expectile and the first-order bridge target  $\psi(\gamma)Q(1-p)$ , for two diagnostic designs separate from the final main grid. In the light-tailed rows, much of the exact-target coverage loss disappears when the bridge target is used. For example, in the  $n = 10000$ ,  $k/n = 0.05$  diagnostic, Burr light moves from about 71% exact coverage to about 92% bridge-target coverage, and light Pareto moves from about 70% to about 95%. In the  $n = 100000$ ,  $k = n^{0.45}$  diagnostic, exact-target coverage for these light-tailed rows is also higher, about 89% for Burr light and 87% for light Pareto. This supports the interpretation that some of the observed undercoverage is a finite-sample bridge effect. The heavy-tailed rows are less improved by this diagnostic, so the conclusion remains cautionary rather than corrective.

This interpretation is consistent with the single-sample extreme-expectile inference evidence of Padoan and Stupfler (2022). There, straightforward Gaussian intervals for quantile-based expectile estimators can be poorly calibrated in finite samples, and the refined log-scale construction keeps bridge-bias and covariance contributions that are asymptotically lower order but still numerically visible. In the present decomposition, the asymptotic argument sends  $B_n(\boldsymbol{\nu})$  and  $C_n$  to zero on the  $\sqrt{k}/\ell$  scale, while the diagnostics show that these terms need not be small at the simulated sample sizes. Extending such corrections to the distributed two-weight estimator would be a separate lower-order inference layer, and may require additional communicated summaries such as local upper order statistics, local tail means, intermediate expectiles, or local estimates of  $\mathbb{E}X$ . The intervals reported here are therefore diagnostics of the first-order plug-in construction, not claims of reliable finite-sample calibration. They should not be recommended without additional calibration in samples of this size.

Table 4.5: Empirical coverage and log-width of nominal 95% intervals in the representative main design. Dashes indicate source-ineligible plug-in constructions, not zero coverage or numerical failure.

| DGP            | $m$ | Regime | Estimator        | Coverage         | Log width        | CI rate          | $\eta_n$ | Bridge gap |
|----------------|-----|--------|------------------|------------------|------------------|------------------|----------|------------|
| Burr heavy     | 10  | strong | Centralised      | 0.755<br>(0.006) | 0.486<br>(0.000) | 1.000<br>(0.000) | 0.051    | 0.009      |
| Burr heavy     | 10  | strong | DPS AMSE plug-in | 0.751<br>(0.006) | 0.498<br>(0.000) | 1.000<br>(0.000) | 0.051    | 0.009      |
| Burr heavy     | 10  | strong | DPS variance     | 0.751<br>(0.006) | 0.498<br>(0.000) | 1.000<br>(0.000) | 0.051    | 0.009      |
| Burr light     | 10  | strong | Centralised      | 0.718<br>(0.006) | 0.202<br>(0.000) | 1.000<br>(0.000) | 0.724    | 0.058      |
| Burr light     | 10  | strong | DPS AMSE plug-in | 0.591<br>(0.007) | 0.208<br>(0.000) | 1.000<br>(0.000) | 0.724    | 0.058      |
| Burr light     | 10  | strong | DPS variance     | 0.591<br>(0.007) | 0.208<br>(0.000) | 1.000<br>(0.000) | 0.724    | 0.058      |
| Pareto heavy   | 10  | strong | Centralised      | 0.770<br>(0.006) | 0.486<br>(0.000) | 1.000<br>(0.000) | 0.051    | 0.012      |
| Pareto light   | 10  | strong | Centralised      | 0.708<br>(0.006) | 0.202<br>(0.000) | 1.000<br>(0.000) | 0.724    | 0.070      |
| Abs. Student-3 | 10  | strong | Centralised      | 0.751<br>(0.006) | 0.270<br>(0.000) | 1.000<br>(0.000) | 0.297    | 0.030      |
| Abs. Student-3 | 10  | strong | DPS AMSE plug-in | 0.920<br>(0.004) | 0.305<br>(0.000) | 1.000<br>(0.000) | 0.297    | 0.030      |
| Abs. Student-3 | 10  | strong | DPS variance     | 0.920<br>(0.004) | 0.305<br>(0.000) | 1.000<br>(0.000) | 0.297    | 0.030      |

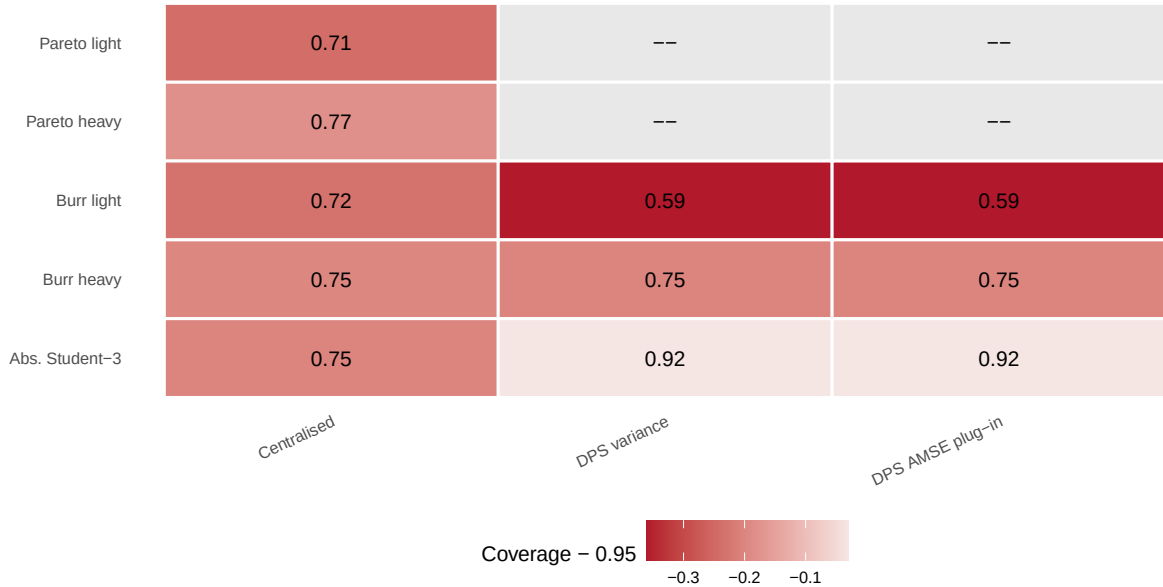


Figure 4.2: Empirical coverage in the representative main design ( $m = 10$ , strong allocation,  $k_j/n_j = 0.05$ ,  $np = 5$ ). Cell labels give coverage; color records the deviation from the nominal 95% level. Cells marked by dashes are source-ineligible exact-Pareto distributed interval rows.

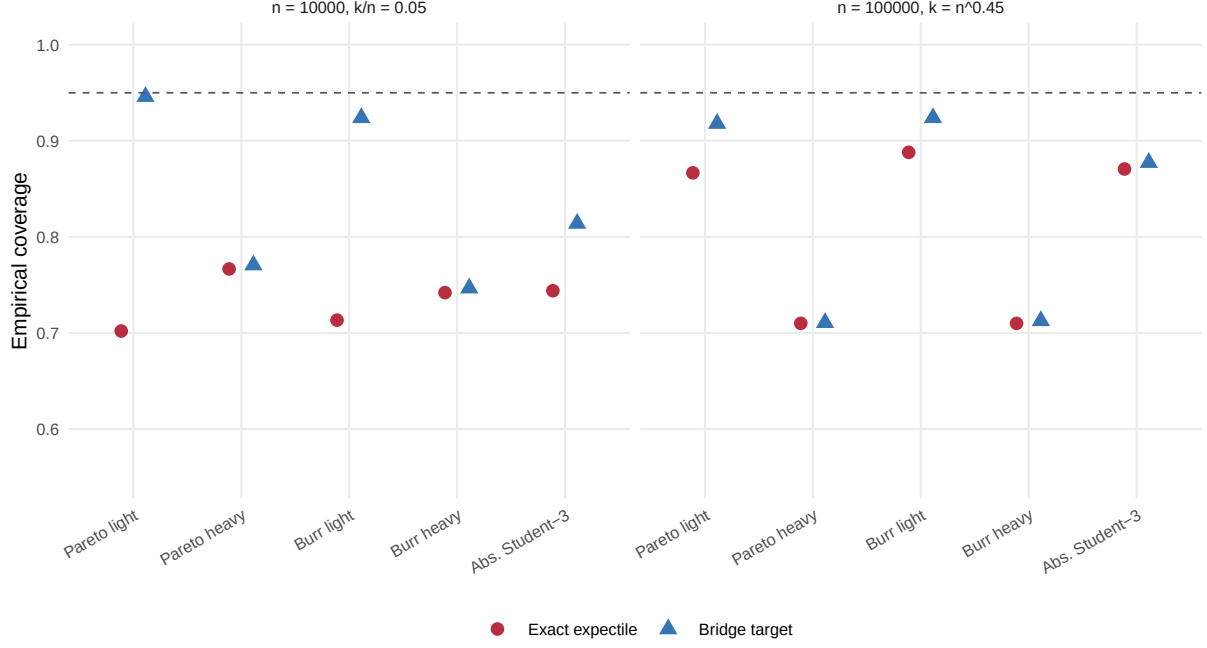


Figure 4.3: Interval diagnostic comparing empirical coverage against the exact population expectile and against the first-order bridge target  $\psi(\gamma)Q(1-p)$ . The diagnostic uses oracle centring and variance objects and is separate from the final main grid.

## 4.6 Decomposition diagnostics

Table 4.6 records the scaled decomposition terms in the representative main design for the estimator using the variance weights of Daouia et al. (2024) with  $\boldsymbol{\nu} = \boldsymbol{\nu}_n^k$ . Here  $A_n$  is the pooled-Weissman term,  $B_n$  is the bridge-Hill term, and  $C_n$  is the population bridge gap. The columns report root mean squared values of  $(\sqrt{k}/\ell)A_n$ ,  $(\sqrt{k}/\ell)B_n$ , and  $(\sqrt{k}/\ell)C_n$ , together with the scaled log error. The final column is the numerical reconstruction check for

$$\log(\hat{\xi}_\tau/\xi_\tau) = A_n(\boldsymbol{\omega}) + B_n(\boldsymbol{\nu}) - C_n.$$

The table answers the same question as the supporting argument: which terms are visible at the pooled-Weissman scale? In the asymptotic argument  $A_n(\boldsymbol{\omega})$  carries the first-order law, while  $B_n(\boldsymbol{\nu})$  and  $C_n$  vanish after multiplication by  $\sqrt{k}/\ell$ . At the simulated sample size, that separation is only partial. The pooled-Weissman component is the largest stochastic component in most rows, but the bridge-Hill term is not negligible in the heavier-tailed cases: its RMS scaled value is 0.474 in Burr heavy and 0.382 in Pareto heavy, compared

with 0.638 and 0.610 for  $A_n(\omega)$ . In contrast,  $B_n(\nu)$  is almost invisible in the light-tailed Pareto and Burr designs.

The population bridge term  $C_n$  shows the opposite pattern. It is small in the heavier-tailed Pareto and Burr cases, but it is comparable to, or larger than, the pooled-Weissman component in the light-tailed cases. This helps explain why point-estimation performance and interval calibration need not move together: some finite-sample error is coming from the population expectile–quantile bridge rather than from the distributed quantile pooling itself. The reconstruction check is at numerical roundoff, confirming that the diagnostics read the exact decomposition rather than an approximate post-hoc split.

Table 4.6: Finite-sample decomposition ledger in the representative main design. The estimator uses the variance pooled-Weissman weights of Daouia et al. (2024) and  $\nu = \nu_n^k$ .

| DGP            | RMS scaled log error | RMS scaled $A_n$ | RMS scaled $B_n$ | scaled $C_n$     | Check          |
|----------------|----------------------|------------------|------------------|------------------|----------------|
| Burr heavy     | 1.080<br>(0.012)     | 0.638<br>(0.007) | 0.474<br>(0.005) | 0.045<br>(0.000) | 0.00<br>(0.00) |
| Burr light     | 0.340<br>(0.003)     | 0.269<br>(0.003) | 0.019<br>(0.000) | 0.279<br>(0.000) | 0.00<br>(0.00) |
| Pareto heavy   | 0.987<br>(0.010)     | 0.610<br>(0.006) | 0.382<br>(0.004) | 0.057<br>(0.000) | 0.00<br>(0.00) |
| Pareto light   | 0.437<br>(0.003)     | 0.255<br>(0.003) | 0.013<br>(0.000) | 0.338<br>(0.000) | 0.00<br>(0.00) |
| Abs. Student-3 | 0.690<br>(0.006)     | 0.589<br>(0.005) | 0.222<br>(0.001) | 0.143<br>(0.000) | 0.00<br>(0.00) |

## 4.7 Sensitivity to thresholds and targets

The threshold sensitivity in Table 4.7 shows that point error and interval calibration can move in different directions. Burr light is the clearest example: increasing  $k_j/n_j$  from 0.03 to 0.10 lowers the log RMSE of the DPS variance-weighted estimator from 0.084 to 0.047, while coverage falls from about 70% to about 32%. In Pareto heavy, where the exact Pareto model removes the Burr-type second-order bias, increasing the threshold improves the corresponding log RMSE from 0.248 to 0.156, but the distributed interval coverage entries are source-ineligible and therefore left blank. In the absolute Student case, the largest threshold worsens log RMSE from 0.116 to 0.285, even though coverage stays near

90%. Threshold choice therefore cannot be defended by a single point-error criterion.

Table 4.8 varies the target through  $np$ . Moving from  $np = 1$  to  $np = 10$  shortens the extrapolation distance, and this lowers log RMSE in Burr heavy and the absolute Student model. It does not uniformly improve calibration. Burr light coverage falls from about 79% at  $np = 1$  to about 43% at  $np = 10$ , Burr heavy coverage falls from about 78% to about 70%, and absolute Student coverage falls from about 92% to about 88%. The target sensitivity again points to an interaction between second-order bias, threshold choice, and first-order interval centring, not to a simple monotone improvement when the target is made less extreme. This suggests that any empirical application should report threshold-sensitivity diagnostics rather than relying on a single threshold fraction.



Table 4.7: Threshold sensitivity for  $m = 10$ , strong allocation and  $np = 5$ .

| DGP            | Threshold rule | Estimator        | $k/n$ | Log RMSE      | Coverage      |
|----------------|----------------|------------------|-------|---------------|---------------|
| Burr heavy     | fraction=0.03  | DPS AMSE plug-in | 0.029 | 0.255 (0.003) | 0.731 (0.006) |
| Burr heavy     | fraction=0.03  | DPS variance     | 0.029 | 0.255 (0.003) | 0.731 (0.006) |
| Burr heavy     | fraction=0.05  | DPS AMSE plug-in | 0.050 | 0.228 (0.002) | 0.732 (0.006) |
| Burr heavy     | fraction=0.05  | DPS variance     | 0.050 | 0.228 (0.002) | 0.733 (0.006) |
| Burr heavy     | fraction=0.1   | DPS AMSE plug-in | 0.100 | 0.260 (0.002) | 0.744 (0.006) |
| Burr heavy     | fraction=0.1   | DPS variance     | 0.100 | 0.260 (0.002) | 0.743 (0.006) |
| Burr light     | fraction=0.03  | DPS AMSE plug-in | 0.029 | 0.084 (0.001) | 0.695 (0.007) |
| Burr light     | fraction=0.03  | DPS variance     | 0.029 | 0.084 (0.001) | 0.696 (0.007) |
| Burr light     | fraction=0.05  | DPS AMSE plug-in | 0.050 | 0.069 (0.001) | 0.580 (0.007) |
| Burr light     | fraction=0.05  | DPS variance     | 0.050 | 0.069 (0.001) | 0.579 (0.007) |
| Burr light     | fraction=0.1   | DPS AMSE plug-in | 0.100 | 0.047 (0.000) | 0.314 (0.007) |
| Burr light     | fraction=0.1   | DPS variance     | 0.100 | 0.047 (0.000) | 0.316 (0.007) |
| Pareto heavy   | fraction=0.03  | DPS AMSE plug-in | 0.029 | —             | —             |
| Pareto heavy   | fraction=0.03  | DPS variance     | 0.029 | 0.248 (0.003) | —             |
| Pareto heavy   | fraction=0.05  | DPS AMSE plug-in | 0.050 | —             | —             |
| Pareto heavy   | fraction=0.05  | DPS variance     | 0.050 | 0.208 (0.002) | —             |
| Pareto heavy   | fraction=0.1   | DPS AMSE plug-in | 0.100 | —             | —             |
| Pareto heavy   | fraction=0.1   | DPS variance     | 0.100 | 0.156 (0.002) | —             |
| Pareto light   | fraction=0.03  | DPS AMSE plug-in | 0.029 | —             | —             |
| Pareto light   | fraction=0.03  | DPS variance     | 0.029 | 0.098 (0.001) | —             |
| Pareto light   | fraction=0.05  | DPS AMSE plug-in | 0.050 | —             | —             |
| Pareto light   | fraction=0.05  | DPS variance     | 0.050 | 0.091 (0.001) | —             |
| Pareto light   | fraction=0.1   | DPS AMSE plug-in | 0.100 | —             | —             |
| Pareto light   | fraction=0.1   | DPS variance     | 0.100 | 0.083 (0.001) | —             |
| Abs. Student-3 | fraction=0.03  | DPS AMSE plug-in | 0.029 | 0.116 (0.001) | 0.895 (0.004) |
| Abs. Student-3 | fraction=0.03  | DPS variance     | 0.029 | 0.116 (0.001) | 0.894 (0.004) |
| Abs. Student-3 | fraction=0.05  | DPS AMSE plug-in | 0.050 | 0.146 (0.001) | 0.917 (0.004) |
| Abs. Student-3 | fraction=0.05  | DPS variance     | 0.050 | 0.146 (0.001) | 0.917 (0.004) |
| Abs. Student-3 | fraction=0.1   | DPS AMSE plug-in | 0.100 | 0.285 (0.001) | 0.905 (0.004) |
| Abs. Student-3 | fraction=0.1   | DPS variance     | 0.100 | 0.285 (0.001) | 0.905 (0.004) |

Table 4.8: Target sensitivity for  $m = 10$ , strong allocation and  $k_j/n_j = 0.05$ .

| DGP            | $np$ | Estimator        | $\eta_n$ | Log RMSE      | Coverage      |
|----------------|------|------------------|----------|---------------|---------------|
| Burr heavy     | 1    | DPS AMSE plug-in | 0.014    | 0.282 (0.003) | 0.779 (0.006) |
| Burr heavy     | 1    | DPS variance     | 0.014    | 0.282 (0.003) | 0.779 (0.006) |
| Burr heavy     | 5    | DPS AMSE plug-in | 0.051    | 0.221 (0.002) | 0.753 (0.006) |
| Burr heavy     | 5    | DPS variance     | 0.051    | 0.221 (0.002) | 0.753 (0.006) |
| Burr heavy     | 10   | DPS AMSE plug-in | 0.090    | 0.207 (0.002) | 0.697 (0.006) |
| Burr heavy     | 10   | DPS variance     | 0.090    | 0.207 (0.002) | 0.698 (0.006) |
| Burr light     | 1    | DPS AMSE plug-in | 0.359    | 0.076 (0.001) | 0.785 (0.006) |
| Burr light     | 1    | DPS variance     | 0.359    | 0.076 (0.001) | 0.785 (0.006) |
| Burr light     | 5    | DPS AMSE plug-in | 0.724    | 0.069 (0.001) | 0.590 (0.007) |
| Burr light     | 5    | DPS variance     | 0.724    | 0.069 (0.001) | 0.590 (0.007) |
| Burr light     | 10   | DPS AMSE plug-in | 1.014    | 0.075 (0.001) | 0.428 (0.007) |
| Burr light     | 10   | DPS variance     | 1.014    | 0.075 (0.001) | 0.427 (0.007) |
| Pareto heavy   | 1    | DPS AMSE plug-in | 0.014    | —             | —             |
| Pareto heavy   | 1    | DPS variance     | 0.014    | 0.245 (0.003) | —             |
| Pareto heavy   | 5    | DPS AMSE plug-in | 0.051    | —             | —             |
| Pareto heavy   | 5    | DPS variance     | 0.051    | 0.204 (0.002) | —             |
| Pareto heavy   | 10   | DPS AMSE plug-in | 0.090    | —             | —             |
| Pareto heavy   | 10   | DPS variance     | 0.090    | 0.190 (0.002) | —             |
| Pareto light   | 1    | DPS AMSE plug-in | 0.359    | —             | —             |
| Pareto light   | 1    | DPS variance     | 0.359    | 0.087 (0.001) | —             |
| Pareto light   | 5    | DPS AMSE plug-in | 0.724    | —             | —             |
| Pareto light   | 5    | DPS variance     | 0.724    | 0.090 (0.001) | —             |
| Pareto light   | 10   | DPS AMSE plug-in | 1.014    | —             | —             |
| Pareto light   | 10   | DPS variance     | 1.014    | 0.098 (0.001) | —             |
| Abs. Student-3 | 1    | DPS AMSE plug-in | 0.128    | 0.222 (0.002) | 0.924 (0.004) |
| Abs. Student-3 | 1    | DPS variance     | 0.128    | 0.222 (0.002) | 0.924 (0.004) |
| Abs. Student-3 | 5    | DPS AMSE plug-in | 0.297    | 0.142 (0.001) | 0.917 (0.004) |
| Abs. Student-3 | 5    | DPS variance     | 0.297    | 0.142 (0.001) | 0.917 (0.004) |
| Abs. Student-3 | 10   | DPS AMSE plug-in | 0.441    | 0.109 (0.001) | 0.880 (0.005) |
| Abs. Student-3 | 10   | DPS variance     | 0.441    | 0.109 (0.001) | 0.880 (0.005) |

## 4.8 Weight stability and limitations

Table 4.9 reports weight diagnostics in the representative main design. The variance row based on Daouia et al. (2024) is the deterministic variance-optimal baseline; there is no separate oracle variance row because, in this common-tail independent design, the oracle and plug-in variance formulas both reduce to  $\omega_j = k_j/k$ . The oracle and plug-in AMSE weights are stable in this equal-threshold grid: negative weights are absent, and the average maximum absolute weight is close to the largest deterministic allocation weight. The diagnostic matters because it rules out an obvious failure mode. The lack of AMSE improvement in Table 4.2 is not caused by unstable AMSE weights; rather, the common-threshold design makes the first-order AMSE criterion reduce to the variance criterion up to an additive bias constant.

Table 4.10 gives the AMSE rule a more discriminating environment. The bridge-Hill vector is fixed at  $\nu_n^k$ , while  $\omega$  varies between the variance, oracle AMSE, and plug-in AMSE weights of Daouia et al. (2024). Here AMSE weighting can matter. In Burr light, the oracle AMSE row raises coverage from 0.600 under variance weights to 0.732, while the plug-in AMSE row raises it to 0.642. In the absolute Student model, AMSE weights reduce log RMSE relative to variance weights, and the plug-in AMSE coverage is close to nominal. This gain comes with a warning: the Student AMSE weights are signed in almost every replication. In Burr heavy, AMSE weights give only a small improvement. Exact Pareto plug-in AMSE rows remain omitted for the source reason described above.

The simulation evidence is intentionally limited. It supports the proposed estimator in the i.i.d. common-marginal distributed setting and suggests that the variance weights of Daouia et al. (2024) are a robust practical baseline for  $\omega$ . Heterogeneous-threshold diagnostics show that AMSE weights are a genuine alternative worth reporting, but not uniformly superior or uniformly stable. The study also shows where the asymptotic theory should not be overinterpreted: first-order plug-in intervals may have empirical coverage below nominal levels at realistic sample sizes, and the exact decomposition reveals finite-sample contributions from terms that are asymptotically negligible under

the Chapter 3 first-order scaling. The study does not address non-identical margins, time series, multivariate targets, random bridge-Hill weights, or lower-order optimality of  $\nu$ . Those questions would require additional mathematical work before they could become simulation targets.

Table 4.9: Pooled-Weissman weight diagnostics in the representative main design.

| DGP            | $m$ | Regime | Estimator        | Neg. rate     | Avg. max $ \omega $ |
|----------------|-----|--------|------------------|---------------|---------------------|
| Burr heavy     | 10  | strong | DPS variance     | 0.000 (0.000) | 0.246 (0.000)       |
| Burr heavy     | 10  | strong | DPS AMSE oracle  | 0.000 (0.000) | 0.246 (0.000)       |
| Burr heavy     | 10  | strong | DPS AMSE plug-in | 0.000 (0.000) | 0.246 (0.000)       |
| Burr light     | 10  | strong | DPS variance     | 0.000 (0.000) | 0.246 (0.000)       |
| Burr light     | 10  | strong | DPS AMSE oracle  | 0.000 (0.000) | 0.246 (0.000)       |
| Burr light     | 10  | strong | DPS AMSE plug-in | 0.000 (0.000) | 0.246 (0.000)       |
| Pareto heavy   | 10  | strong | DPS variance     | 0.000 (0.000) | 0.246 (0.000)       |
| Pareto heavy   | 10  | strong | DPS AMSE oracle  | 0.000 (0.000) | 0.246 (0.000)       |
| Pareto heavy   | 10  | strong | DPS AMSE plug-in | —             | —                   |
| Pareto light   | 10  | strong | DPS variance     | 0.000 (0.000) | 0.246 (0.000)       |
| Pareto light   | 10  | strong | DPS AMSE oracle  | 0.000 (0.000) | 0.246 (0.000)       |
| Pareto light   | 10  | strong | DPS AMSE plug-in | —             | —                   |
| Abs. Student-3 | 10  | strong | DPS variance     | 0.000 (0.000) | 0.246 (0.000)       |
| Abs. Student-3 | 10  | strong | DPS AMSE oracle  | 0.000 (0.000) | 0.241 (0.000)       |
| Abs. Student-3 | 10  | strong | DPS AMSE plug-in | 0.000 (0.000) | 0.244 (0.000)       |

Table 4.10: Pooled-Weissman weight comparison under heterogeneous local threshold fractions. The bridge-Hill weights are fixed at  $\boldsymbol{\nu} = \boldsymbol{\nu}_n^k$ .

| DGP            | Estimator        | Log bias          | Log RMSE         | Coverage         | Neg. rate        | Avg. max $ \omega $ |
|----------------|------------------|-------------------|------------------|------------------|------------------|---------------------|
| Burr heavy     | DPS AMSE oracle  | 0.094<br>(0.003)  | 0.231<br>(0.002) | 0.755<br>(0.006) | 0.000<br>(0.000) | 0.174<br>(0.000)    |
| Burr heavy     | DPS AMSE plug-in | 0.091<br>(0.003)  | 0.231<br>(0.002) | 0.767<br>(0.006) | 0.000<br>(0.000) | 0.203<br>(0.000)    |
| Burr heavy     | DPS variance     | 0.097<br>(0.003)  | 0.234<br>(0.002) | 0.744<br>(0.006) | 0.000<br>(0.000) | 0.147<br>(0.000)    |
| Burr light     | DPS AMSE oracle  | -0.036<br>(0.001) | 0.067<br>(0.001) | 0.732<br>(0.006) | 0.000<br>(0.000) | 0.174<br>(0.000)    |
| Burr light     | DPS AMSE plug-in | -0.037<br>(0.001) | 0.068<br>(0.001) | 0.642<br>(0.007) | 0.000<br>(0.000) | 0.203<br>(0.000)    |
| Burr light     | DPS variance     | -0.034<br>(0.001) | 0.067<br>(0.001) | 0.600<br>(0.007) | 0.000<br>(0.000) | 0.147<br>(0.000)    |
| Pareto heavy   | DPS AMSE oracle  | -0.010<br>(0.003) | 0.208<br>(0.002) | 0.746<br>(0.006) | 0.000<br>(0.000) | 0.147<br>(0.000)    |
| Pareto heavy   | DPS AMSE plug-in | —                 | —                | —                | —                | —                   |
| Pareto heavy   | DPS variance     | -0.010<br>(0.003) | 0.208<br>(0.002) | —<br>(0.000)     | 0.000<br>(0.000) | 0.147<br>(0.000)    |
| Pareto light   | DPS AMSE oracle  | -0.072<br>(0.001) | 0.091<br>(0.001) | 0.687<br>(0.007) | 0.000<br>(0.000) | 0.147<br>(0.000)    |
| Pareto light   | DPS AMSE plug-in | —                 | —                | —                | —                | —                   |
| Pareto light   | DPS variance     | -0.072<br>(0.001) | 0.091<br>(0.001) | —<br>(0.000)     | 0.000<br>(0.000) | 0.147<br>(0.000)    |
| Abs. Student-3 | DPS AMSE oracle  | 0.068<br>(0.002)  | 0.144<br>(0.001) | 0.876<br>(0.005) | 1.000<br>(0.000) | 0.409<br>(0.000)    |
| Abs. Student-3 | DPS AMSE plug-in | 0.103<br>(0.001)  | 0.142<br>(0.001) | 0.945<br>(0.003) | 0.996<br>(0.001) | 0.264<br>(0.000)    |
| Abs. Student-3 | DPS variance     | 0.132<br>(0.001)  | 0.162<br>(0.001) | 0.923<br>(0.004) | 0.000<br>(0.000) | 0.147<br>(0.000)    |

# Chapter 5

## Conclusion

This work has studied inference for extreme expectile-based risk measures in a distributed heavy-tailed setting. The construction builds on the optimal-pooling framework of Daouia et al. (2024): it first forms a geometrically pooled Weissman estimator of an extreme quantile and only then applies the high-expectile/high-quantile bridge. The two-weight formulation separates two roles that are easy to conflate:  $\omega$  controls the pooled-Weissman component, while  $\nu$  enters only through the Hill estimator used in the outer bridge. This separation makes clear which part of the estimator is governed by the pooling theory of Daouia et al. (2024).

### 5.1 Main findings

The theoretical argument is useful precisely because it is explicit about what has to be controlled. The log error splits into the pooled-Weissman error, the plug-in error from evaluating  $\psi$  at a pooled Hill estimator, and the population error in replacing the expectile by the asymptotic quantile bridge. In the i.i.d. common-marginal distributed regime, under the extreme-Weissman conditions and the bridge-rate condition, the Chapter 3 asymptotic argument treats the two bridge terms as lower order on the pooled-Weissman scale. The resulting conjectural limiting law therefore depends on  $\omega$ , but not on  $\nu$ .

This is the main conjectural message of [Chapter 3](#): first-order variance and AMSE optimisation transfer to  $\omega$  from the theory of Daouia et al. (2024), while  $\nu$  is first-order unidentified in this regime. The simulations then adopt the fixed threshold-allocation bridge vector used in the plug-in construction.

This work then formulates a conditional transfer to estimated  $\omega$  weights satisfying the random-weight conditions of Daouia et al. (2024, Supplementary Material, Theorem A.1, Theorem 2 and Corollary 8). This transfer is deliberately conditional: when the corresponding source construction supplies consistent estimated weights and consistent plug-in centring and variance objects, the expectile estimator is expected to inherit the same first-order studentised limit. The resulting confidence interval is obtained by inverting the transferred log-scale expectile statistic. It is not a direct reuse of a quantile interval formula, and it does not introduce an expectile-specific plug-in bias or variance estimator.

## 5.2 Finite-sample lessons

The simulation study in [Chapter 4](#) is matched to the theoretical scope of the Chapter 3 analysis. It uses the practical bridge-Hill convention and varies only the pooled-Weissman weights. Within this design, the variance weights of Daouia et al. (2024) provide a stable practical baseline. In the representative imbalanced grid, they make the distributed estimator track the centralised benchmark closely for point estimation, whereas equal pooled-Weissman weights are less efficient under strong imbalance.

The finite-sample role of AMSE weights is more conditional. In the main equal-threshold grid, oracle and plug-in AMSE weights are nearly indistinguishable from variance weights. Under heterogeneous local threshold fractions, AMSE weights can matter more, and in some rows they improve either coverage or log RMSE. They are not uniformly superior, however, and the diagnostics show that signed or unstable weights can appear. Thus the simulation evidence supports reporting AMSE weights as a meaningful source-based alternative, not as a uniformly preferable default.

The interval results are more cautionary. The first-order log-scale intervals have empirical coverage below nominal levels in several finite-sample designs, and the centralised oracle benchmark does too. The decomposition diagnostics explain why this is not only a distributed-splitting issue. At the simulated sample sizes, the terms  $B_n(\boldsymbol{\nu})$  and  $C_n$ , although asymptotically negligible under the Chapter 3 first-order scaling, can still be visible. In particular, finite-sample bridge error can affect calibration even when point estimation is competitive. The interval study should therefore be read as a diagnostic of the plug-in construction rather than as evidence of strong finite-sample nominal coverage.

### 5.3 Limitations and future work

The limitations are mainly in scope. The analysis is carried out in the finite- $m$  i.i.d. common-marginal distributed setting. This restriction is useful: it keeps the target expectile unambiguous and lets the pooled-Weissman theory of Daouia et al. (2024) be combined with the high-expectile/high-quantile bridge without adding a separate target-selection problem. A natural next step is to move beyond common margins, for example to tail-homoskedastic or otherwise heterogeneous sources. That extension would first have to specify the common object being estimated and then revisit the decomposition terms source by source, because the population bridge and the centring objects would no longer be identical across machines.

A second limitation is that the present argument is first-order. Under the bridge-rate condition used in Chapter 3, the terms coming from the plug-in  $\psi$  factor and from the population expectile–quantile bridge are lower order on the pooled-Weissman scale. This is precisely what allows the pooling-weight criteria of Daouia et al. (2024, Corollary 5) to apply to  $\boldsymbol{\omega}$  at first order, but it also means that the bridge-Hill vector  $\boldsymbol{\nu}$  is not selected by the first-order criterion. A refined analysis retaining lower-order bridge terms would be needed before any performance-based data-adaptive rule for  $\boldsymbol{\nu}$  could be justified.

The same boundary applies to estimated weights and plug-in inference. The estimated-weight transfer is an  $\boldsymbol{\omega}$ -transfer: when the plug-in construction of Daouia et al. (2024,



§2.2 and Corollary 1, Corollaries 6–7) supplies consistent estimated weights and consistent plug-in centring and variance objects, the pooled extreme expectile estimator inherits the corresponding first-order log-scale limit. The bridge-Hill convention remains deterministic here. Allowing a random or fully data-adaptive bridge-Hill vector would require checking that the compact-event Taylor argument for the  $\psi$  term still gives a negligible contribution at the relevant scale, and, if lower-order accuracy is desired, that the retained bridge terms are correctly represented.

The finite-sample interval results point to a related direction rather than to a purely numerical gap. The simulations show that first-order log-scale plug-in intervals can have empirical coverage below nominal levels, including for the centralised benchmark. This suggests that calibration is affected by threshold choice, second-order tail bias and finite-sample bridge effects that are deliberately suppressed by the first-order theory. More systematic calibration work would therefore be most informative after the relevant lower-order terms or broader source settings have been theoretically isolated. Possible extensions include bias-corrected centring, alternative studentisation, threshold diagnostics, intervals retaining an estimable population bridge term, and simulation-guided sensitivity analysis. Each would need to be tied to a specific extension of the asymptotic argument.

The simulation study is narrow for the same reason. It is designed to test the promoted estimator, the variance and AMSE choices for  $\omega$  from Daouia et al. (2024), the fixed bridge-Hill convention, and the associated plug-in construction in a controlled i.i.d. common-marginal setting. Broader simulation grids, real-data examples, time-series dependence, non-identical sources, growing numbers of machines and more systematic threshold-selection rules are all natural next analyses. Their interpretation, however, depends on which theoretical extension is being assessed. Simulating such designs before the corresponding target and decomposition have been fixed would risk turning numerical evidence into a substitute for the missing mathematical layer.

Finally, this work follows the geometrically pooled Weissman construction because it is directly supported by the pooled-Weissman framework of Daouia et al. (2024, Theorem 2

and Corollary 8) and by the log-scale decomposition used in [Chapter 3](#). Other architectures remain possible. One could transmit and pool local quantile-based or LAWS expectile estimators, compare geometric and arithmetic pooling of expectile estimates, or change the communication protocol to send additional summaries such as local upper order statistics, local tail means, intermediate expectiles, or local estimates of  $\mathbb{E} X$ . These are not merely cosmetic variants of the estimator studied here. They would define different statistical objects, and should therefore be analysed through their own exact decompositions before being compared theoretically or in finite samples.

# Bibliography

- Bellini, F., B. Klar, A. Müller, and E. Rosazza Gianin (2014). “Generalized quantiles as risk measures”. In: *Insurance: Mathematics and Economics* 54, pp. 41–48.
- Belzile, L., B. G. Manjunath, and F. Caeiro (2026). *evt0: Mean of Order P, Peaks over Random Threshold Hill and High Quantile Estimates*. R package version 1.1.5, accessed 2026-06-02. URL: <https://cran.r-universe.dev/evt0/evt0.pdf>.
- Daouia, A., S. Girard, and G. Stupfler (2018). “Estimation of tail risk based on extreme expectiles”. In: *Journal of the Royal Statistical Society. Series B* 80.2, pp. 263–292. DOI: [10.1111/rssb.12254](https://doi.org/10.1111/rssb.12254).
- (2019). “Extreme M-quantiles as risk measures: From  $L^1$  to  $L^p$  optimization”. In: *Bernoulli* 25.1, pp. 264–309. DOI: [10.3150/17-BEJ987](https://doi.org/10.3150/17-BEJ987).
- (2020). “Tail expectile process and risk assessment”. In: *Bernoulli* 26.1, pp. 531–556. DOI: [10.3150/19-BEJ1137](https://doi.org/10.3150/19-BEJ1137).
- Daouia, A., S. A. Padoan, and G. Stupfler (2024). “Optimal weighted pooling for inference about the tail index and extreme quantiles”. In: *Bernoulli* 30.2, pp. 1287–1312. DOI: [10.3150/23-BEJ1632](https://doi.org/10.3150/23-BEJ1632). arXiv: [2111.03173](https://arxiv.org/abs/2111.03173).
- Davison, A. C., S. A. Padoan, and G. Stupfler (2023). “Tail risk inference via expectiles in heavy-tailed time series”. In: *Journal of Business & Economic Statistics* 41.3, pp. 876–889. DOI: [10.1080/07350015.2022.2078332](https://doi.org/10.1080/07350015.2022.2078332).
- de Haan, L. and A. Ferreira (2006). *Extreme Value Theory: An Introduction*. Springer Series in Operations Research and Financial Engineering. New York: Springer.

- Ehm, W., T. Gneiting, A. Jordan, and F. Krüger (2016). “Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings”. In: *Journal of the Royal Statistical Society. Series B* 78.3, pp. 505–562.
- Gneiting, T. (2011). “Making and evaluating point forecasts”. In: *Journal of the American Statistical Association* 106.494, pp. 746–762.
- Hill, B. M. (1975). “A simple general approach to inference about the tail of a distribution”. In: *The Annals of Statistics* 3.5, pp. 1163–1174.
- Newey, W. K. and J. L. Powell (1987). “Asymmetric least squares estimation and testing”. In: *Econometrica* 55.4, pp. 819–847.
- Padoan, S. A. (2026). *ExtremeRisks: Extreme Risk Measures*. R package version 0.0.5, accessed 2026-06-02. URL: <https://cran.r-universe.dev/ExtremeRisks/doc/manual.html>.
- Padoan, S. A. and G. Stupfler (2022). “Joint inference on extreme expectiles for multivariate heavy-tailed distributions”. In: *Bernoulli* 28.2, pp. 1021–1048. DOI: [10.3150/21-BEJ1375](https://doi.org/10.3150/21-BEJ1375).
- Weissman, I. (1978). “Estimation of parameters and large quantiles based on the  $k$  largest observations”. In: *Journal of the American Statistical Association* 73.364, pp. 812–815.
- Ziegel, J. F. (2016). “Coherence and elicibility”. In: *Mathematical Finance* 26.4, pp. 901–918. DOI: [10.1111/mafi.12080](https://doi.org/10.1111/mafi.12080).